

Tessera POS: system inventory (approved)

Compiled 27 September 2026 from the code (menus, screens, forms, permissions) and checked against the running system (menus fetched while signed in as Owner, Branch Manager, Accountant and Storekeeper; reports listed by the running system). This file is the working checklist for the user guide. It is not part of the client guide.

Role key:

Code	Role
Ow	Owner
Ad	Admin
BM	Branch Manager
Ca	Cashier
SK	Storekeeper (the brief's "stock clerk")
Ac	Accountant
TS	Tessera Support (Tessera's own staff, not the client)

"See" means the menu item appears. Where roles can see a screen but not do everything on it, the Purpose column says who can do what.

How the modules connect

The main steps (purple, and the till in amber) follow a business day from left to right. The white boxes are the records each step updates. Dotted arrows show what feeds the dashboard, reports and alerts. The source file is `diagrams/modules-flow.mmd` (Mermaid), so the chart can be edited and redrawn.

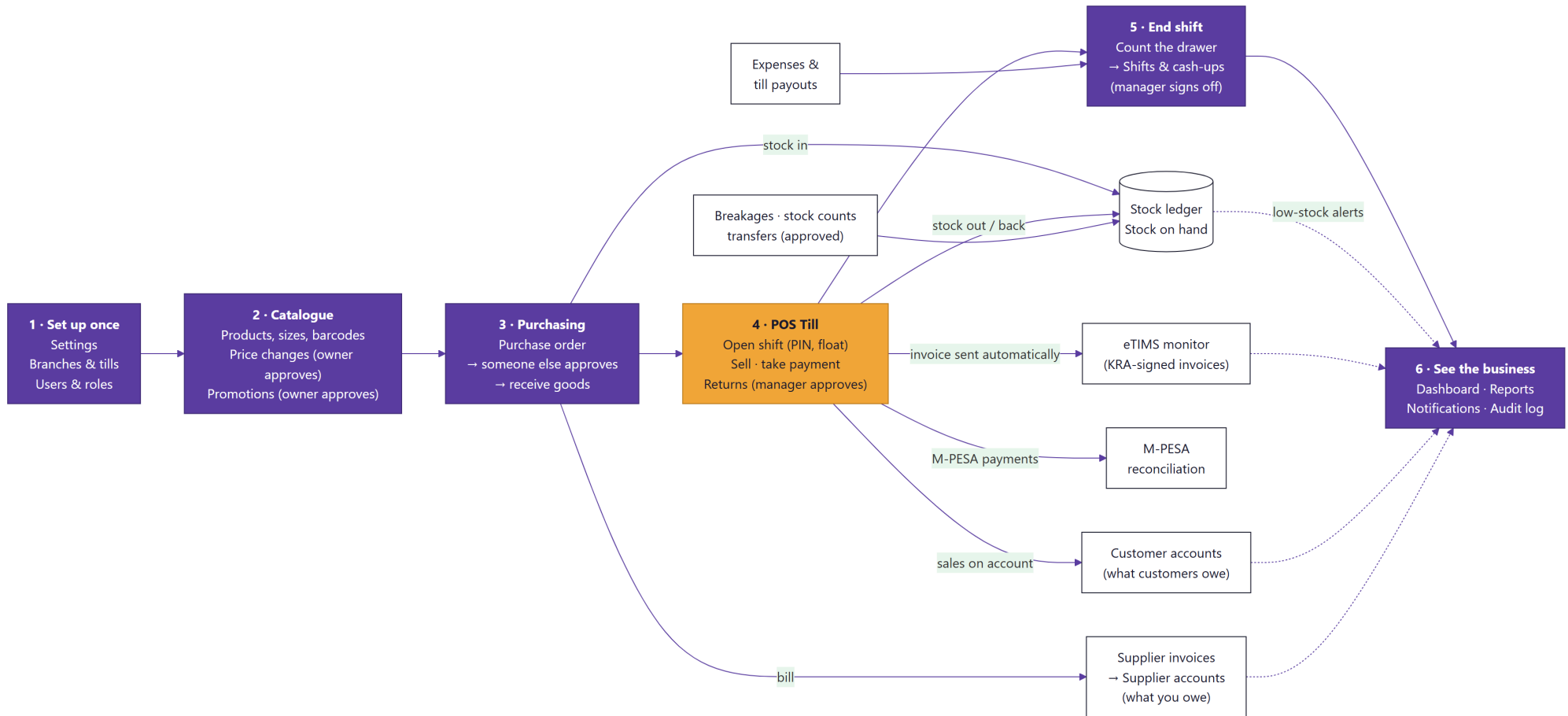


Figure 1: How the Tessa POS modules connect, in the order of a business day

- 1. Set up once:** the owner fills in Settings, adds branches and tills, and creates users with their roles and till PINs.
- 2. Catalogue:** products, sizes and barcodes are added. New prices and promotions wait for the owner's approval.
- 3. Purchasing:** a purchase order is raised and approved by someone else, then the goods are received. Stock goes up in the stock ledger, and the supplier's bill goes to Supplier accounts.
- 4. POS Till:** the cashier signs in with a PIN and opens the shift. Each sale lowers the stock, and its tax invoice is sent to KRA automatically (eTIMS monitor). M-PESA payments go to M-PESA reconciliation and sales on account to Customer accounts. Returns need a manager and put stock back.
- 5. End shift:** the cashier counts the drawer. Payouts and expenses are taken into account, and a manager signs off the cash-up.
- 6. See the business:** the dashboard, reports, notifications (low stock, licences, daily summary) and the audit log bring it all together.

1. Signing in and the account menu (no menu item)

Module	Menu	Screen / form	Roles	Purpose
Sign in	—	Back-office sign-in (email or phone + password, "Keep me signed in")	All except Ca	Sign in to the back office.
Sign in	—	Forgot password?	All	Explains that the owner or an administrator resets passwords. There is no self-service reset by email or SMS.
Sign in	—	Two-step login set-up (QR code, 6-digit code, recovery codes)	Ow, Ad, TS (TS always)	Extra security for owners and admins. The owner can switch it off under Settings.
Sign in	—	Change password (forced when expired or first sign-in)	All back-office users	Change your own password.
Account menu (top right)	—	Open till screen · Change password · Two-step login · Sign out	All back-office users	Personal actions.
Notifications	—	Bell with unread alerts	All back-office users	Alerts: low stock, licence expiry, large refunds, cash variance, eTIMS failures, daily summary.
Session security	—	Signed out after 30 minutes idle (back office)	All back-office users	Protects an unattended computer.

2. The till (full-screen, opened from POS Till or /till)

Module	Menu	Screen / form	Roles	Purpose
POS Till	POS Till	Till launcher page	Ow, Ad, BM, Ca, TS	Opens the full-screen till on this computer.
POS Till	—	Set up this device as a till (branch, till name, location, opening float)	Ow, Ad, TS (manager sign-in)	One-time pairing of a computer as a till.
POS Till	—	Who's on the till? (choose name, PIN pad, start shift)	Ca, BM, Ow... (anyone with a PIN and selling rights)	Cashier sign-in by PIN; starts or resumes the shift.
POS Till	—	Selling screen: search/scan, favourites (best sellers), cart, quantity, customer, totals, Pay	Ca and above	Making sales.
POS Till	—	Line edit (quantity, discount, price change) + Manager approval (manager PIN)	Ca (within limit); manager approves beyond	Discounts, price overrides, removing items.
POS Till	—	Age check	Ca	Confirm the customer is 18 or over (setting).

Module	Menu	Screen / form	Roles	Purpose
POS Till	—	Take payment: M-PESA (payment request to phone, or pick a payment already made), Cash, Card, On account, Split; customer KRA PIN	Ca	Taking payment.
POS Till	—	Sale complete (change due, receipt preview, Print receipt, Next customer)	Ca	Finish the sale.
POS Till	—	Customer picker	Ca	Attach a registered customer (wholesale price, on-account, tax invoice).
POS Till	—	Park sale / Recall parked sales	Ca	Hold a sale while serving someone else.
POS Till	—	Sell by tot (open bottles)	Ca	Pour and sell tots from an open bottle (setting).
POS Till	—	Price check	Ca	Look up a price without selling.
POS Till	—	Return or reprint (find receipt, pick items, reason, refund)	Ca requests; manager approves (setting)	Customer returns and receipt reprints.
POS Till	—	Cash drop to the safe	Ca	Move cash from the drawer to the safe during a shift.
POS Till	—	Pay out cash for an expense (manager witnesses by PIN)	Ca + manager	Small expenses paid from the drawer.
POS Till	—	Lock screen (manual, or automatic after 2 minutes idle; sale kept)	Ca; manager can unlock	Leave the till safely.
POS Till	—	Switch cashier	Ca	Hand over without ending the shift.
POS Till	—	End shift: count the cash drawer (notes and coins) → Shift ended (expected, counted, variance, reason)	Ca	Cash-up at the end of a shift.
POS Till	—	Offline mode banner (still selling; cash and card only; sends when back online)	Ca	Keep selling when the internet is down.
POS Till	—	Licensed-hours banner (no alcohol outside licensed hours)	Ca	Shown when the owner has switched on the licensed-hours lock.

3. Back-office menus

Module	Menu	Screen / form	Roles (see)	Purpose
Dashboard	Dashboard	KPI tiles, Needs attention, Shifts open now, Low stock, Best sellers/slow movers, Branch comparison, Cash variance by cashier, and others	Ow, Ad, BM, SK, Ac, TS	Today's business at a glance. Tiles depend on the role and the settings. The storekeeper's version shows stock tiles only and is their home screen.

Module	Menu	Screen / form	Roles (see)	Purpose
Sales & Shifts	Sales	Sales list → Sale detail (lines, payments, receipt, returns)	Ow, Ad, BM, Ac, TS	Find and inspect any sale; reprint.
Sales & Shifts	Shifts & cash-ups	Shift list → sign off a cash-up (with cashier's reason)	Ow, Ad, BM, TS	Managers check and sign off every cash-up.
Catalogue	Products	Product list → New/Edit product, Sizes (add size), Add barcode, Add pack (e.g. crate of 12), Change price, Price history, Product detail tab	See: Ow, Ad, BM, SK, Ac, TS. Edit: Ow, Ad, TS	The items the shop sells.
Catalogue	Price changes	Pending price changes → Approve / Reject	Ow, Ad, TS (Ow and TS approve)	Maker-checker for prices.
Catalogue	Promotions	Promotion list, New promotion (percent/amount off, minimum quantity, dates, happy-hour times, categories/items), approve, discount given	Ow, Ad, BM, TS (Ow and TS approve)	Automatic discounts at the till.
Catalogue	Brands & categories	Add/Edit brand, Add/Edit category	Ow, Ad, TS	Organise products.
Inventory	Stock on hand	Stock per item and branch; reorder level per item	Ow, Ad, BM, SK, Ac, TS	What is in stock where.
Inventory	Breakages & adjustments	New stock adjustment (breakage, theft, expiry, damage, opening stock...) → Approve / Reject	See: Ow, Ad, BM, SK, Ac, TS. Record: SK, Ow, Ad. Approve: BM, Ow, Ad	Stock losses and corrections.
Inventory	Transfers	Request a transfer → Approve → Dispatch → Receive; Cancel	See: as above. Request: SK, Ow, Ad. Approve: BM, Ow, Ad	Move stock between branches.
Inventory	Stock counts	Start a stock count (blind count option) → Count sheet → Submit → Approve / Reject	See: as above. Count: SK, Ow, Ad. Approve: BM, Ow, Ad	Physical stock-take.
Inventory	Open bottles (tots)	Open bottles per branch, remaining ml, Write off	Ow, Ad, BM, SK, Ac, TS	Track bottles being sold by the tot.
Inventory	Stock ledger	Every stock movement with its document	Ow, Ad, BM, SK, Ac, TS	The full history of stock.
Purchasing	Purchase orders	New purchase order → Approve / Reject / Cancel → Receive goods (received, damaged) → Goods-received note; Purchase order detail tab	See: Ow, Ad, BM, SK, Ac, TS. Raise: Ow, Ad, Ac. Approve: BM, Ow, Ad (never their own). Receive: SK, BM, Ow, Ad	Order from suppliers and receive deliveries.
Purchasing	Supplier invoices	Record supplier invoice (matched to order and delivery)	See: as above. Record: Ow, Ad, Ac	Bills from suppliers.
Purchasing	Returns to supplier	Return to supplier → Approve / Reject → Credit note	See: as above	Send damaged or wrong goods back.
Purchasing	Suppliers	Supplier list → Add/Edit supplier	See: as above. Edit: Ow, Ad, Ac	Supplier details and terms.

Module	Menu	Screen / form	Roles (see)	Purpose
Purchasing	Supplier accounts	Payables aging, Payment to supplier, Reverse payment, Statement	Ow, Ad, BM, Ac, TS (pay: Ow, Ad, Ac)	What the shop owes each supplier.
Payments	M-PESA reconciliation	M-PESA payments vs sales, unmatched payments, Match to a sale	Ow, Ad, BM, Ac, TS	Make sure every M-PESA payment belongs to a sale.
Customers	Customers	Customer list → Register/Edit customer, Customer detail tab, Credit account (limit, terms), Anonymise	See: Ow, Ad, BM, Ca, Ac, TS. Add/Edit: Ow, Ad, BM. Credit: Ow, Ad, Ac. Anonymise: Ow, Ad. Cashiers: search and select only (no balances); adding customers is a fix needed (internal notes)	Registered customers (businesses, wholesale, account customers).
Customers	Customer accounts	Receivables aging, Payment from customer, Statement	Ow, Ad, BM, Ac, TS	What customers owe the shop.
Expenses	Expenses	Expense list and totals, Record an expense, Approve / Reject, Reverse	Ow, Ad, BM, Ac, TS (approve: BM, Ow, Ad; never their own)	Petty cash, bank and M-PESA spending.
Compliance	eTIMS monitor	Signed / pending / refused invoices, daily check, retry	Ow, Ad, BM, Ac, TS	KRA eTIMS status of every sale and return.
Compliance	Licences & permits	Licence register, add, Renew, Remove, print on receipt	See: Ow, Ad, BM, Ac, TS. Manage: Ow, Ad, TS	Licence expiry reminders.
Reports	Reports	Report catalogue (22 entries) → each report with filters, CSV export and print	Ow, Ad, BM, Ac, TS (profit and financial reports need the financial permission)	See the list in section 4.
Administration	Branches	Branches, Tills per branch, Disconnect a till	Ow, Ad, TS	Branches and till devices.
Administration	Users & roles	User list → Add/Edit user (role, branches), Till PIN, Reset two-step login	Ow, Ad, TS	Staff accounts and access.
Administration	Settings	Nine sections: Business, Branding, Receipts, Sales screen, Products and stock, Payments, Staff and roles, Notifications, Integrations. Also: settings per business, branch or till; Industry preset; History and Undo; "Locked after first use" list	Ow, Ad, TS (BM: branch-level settings only)	How the system looks and works. Settings for unfinished features are hidden (see internal notes).
Administration	Messages sent	Log of SMS / WhatsApp / in-app alerts sent	Ow, Ad, TS	Check what alerts went out (SMS runs in demo mode).
Administration	Audit log	Who did what and when, with filters	Ow, Ad, BM, Ac, TS	Accountability and investigations.

4. Reports (as listed by the running system)

Group	Report
Sales	Sales summary
Sales	Sales by item, product, category or brand
Sales	Sales by cashier, branch or payment method
Sales	Voids, discounts & price overrides
Sales	Returns & refunds register
Inventory	Current stock by branch
Inventory	Stock valuation as at a date
Inventory	Adjustments & losses by reason
Inventory	Stock count variance
Inventory	Transfers & discrepancies
Inventory	Low stock & reorder suggestions
Inventory	Stock movement ledger (links to Inventory → Stock ledger)
Financial	Revenue, cost of goods & gross profit
Financial	Shift cash-up & variance
Financial	Purchases by supplier
Financial	Receivables aging (customer accounts)
Financial	Payables aging (supplier balances)
Financial	Expenses (petty cash)
Financial	M-PESA reconciliation (links to Payments)
Compliance	eTIMS status, failures & daily check (links to Compliance)
Compliance	Licence & permit expiry list (links to Compliance)
Compliance	Audit trail (links to Administration → Audit log)

5. Screens (screenshots from the running system)

How these were taken:

- **Back office (Figures 2 to 35):** 1440×900, signed in as the owner on the local demo system, just looking (nothing was changed).
- **Till (Figures 36 to 53):** 1366×768, on the separate test system, with a cashier (Otieno) making a real test sale.

The small "N" badge in the bottom-left corner appears only in development mode. The final guide will be shot without it, on the "Karen Wines & Spirits" demo data once you approve it.

5.1 Back office

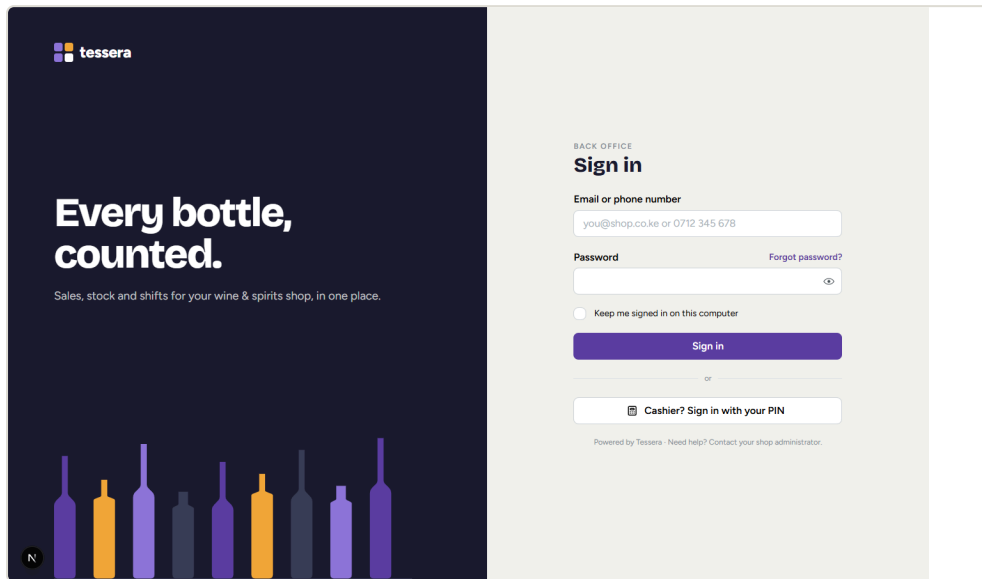


Figure 2: Back-office sign-in

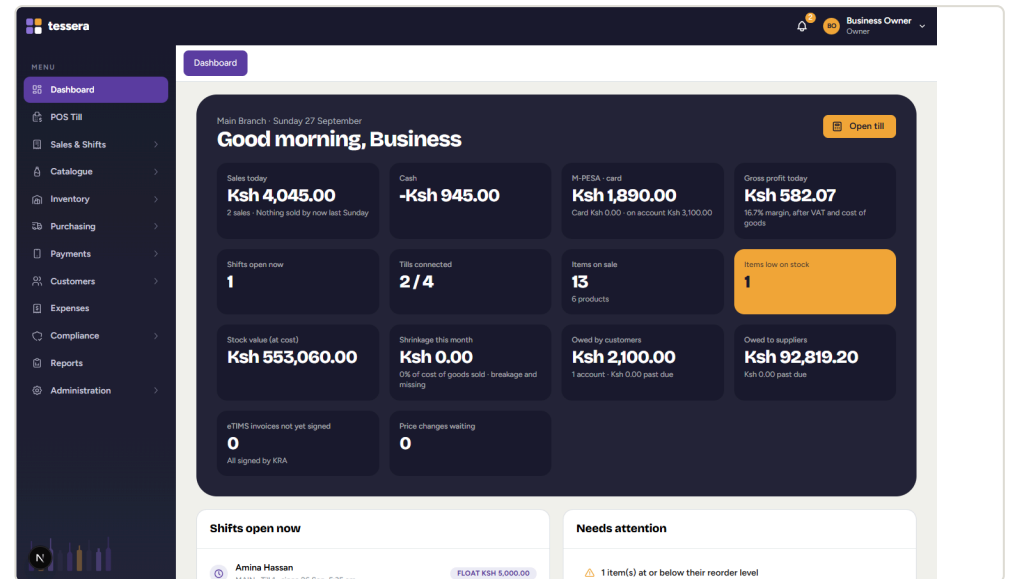


Figure 3: Dashboard (owner view)

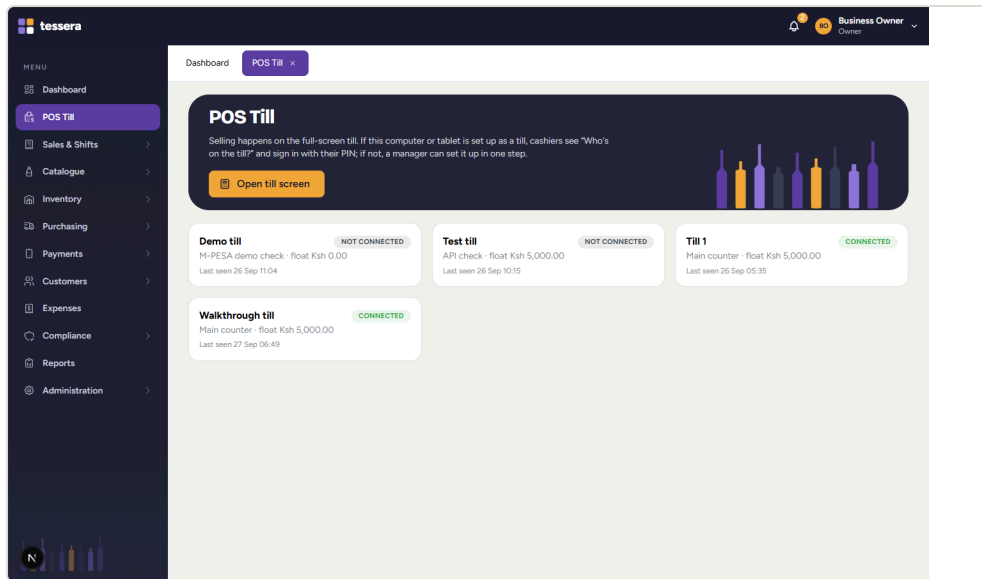


Figure 4: POS Till: opens the till screen on this computer

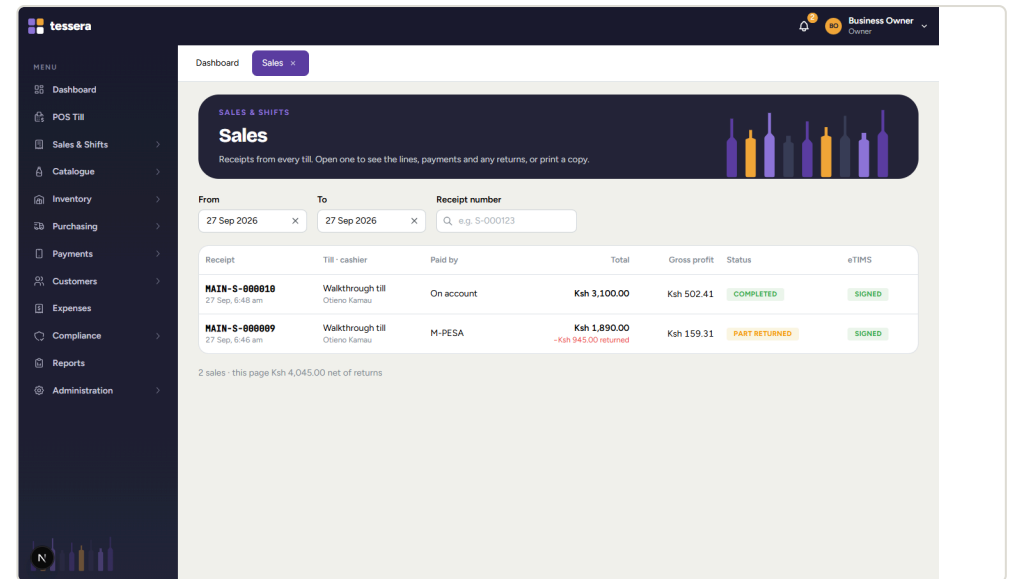


Figure 5: Sales & Shifts → Sales

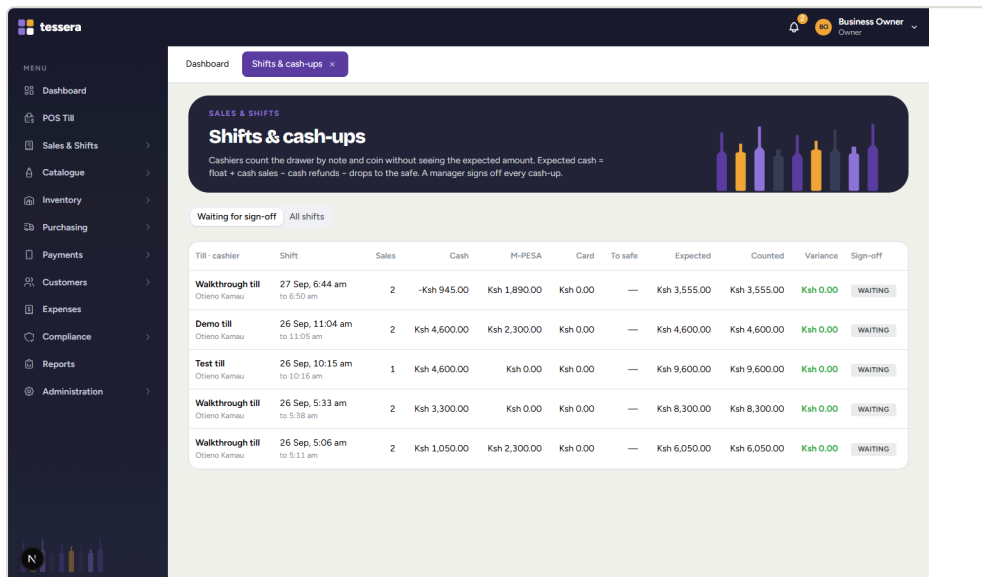


Figure 6: Sales & Shifts → Shifts & cash-ups

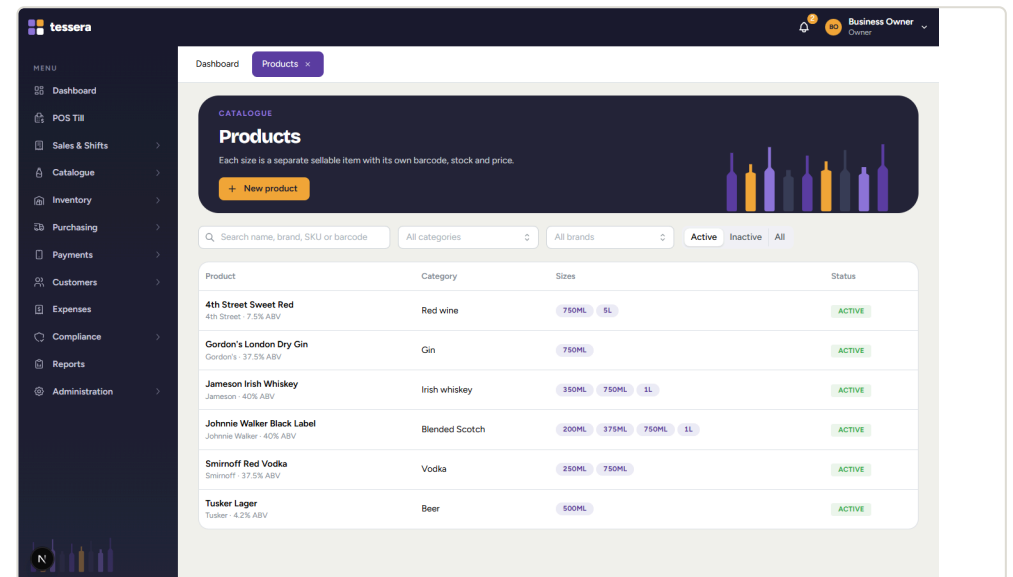


Figure 7: Catalogue → Products

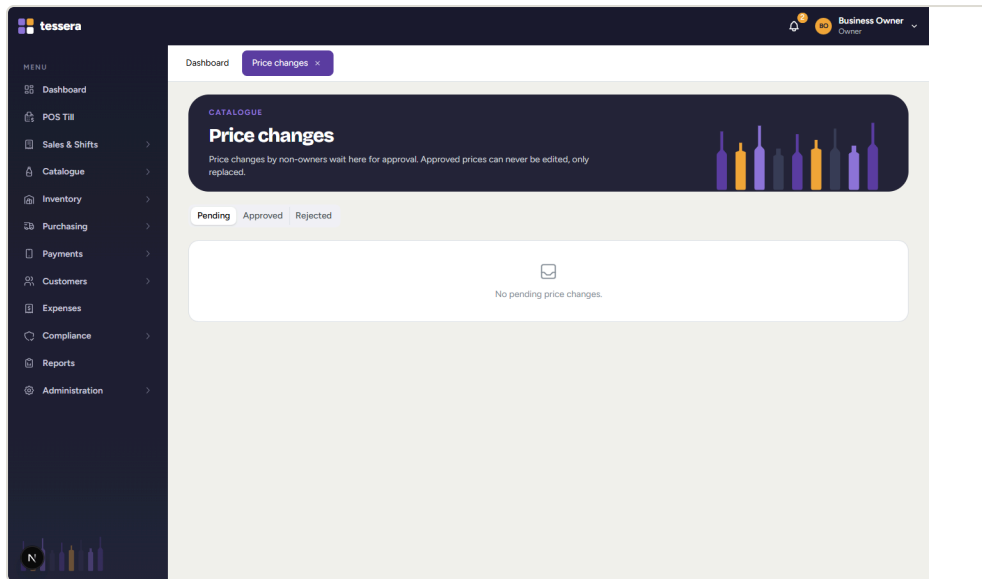


Figure 8: Catalogue → Price changes

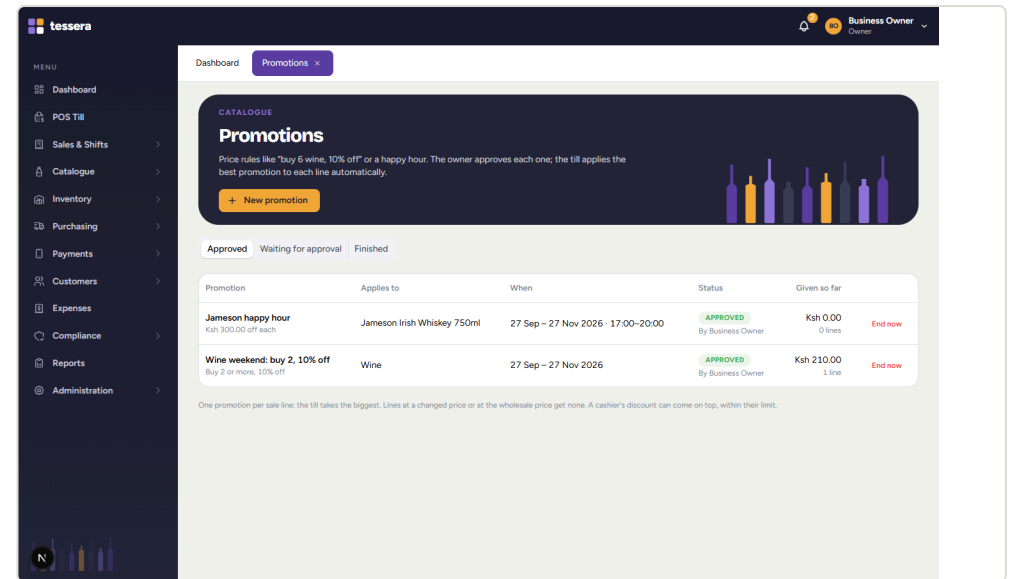


Figure 9: Catalogue → Promotions

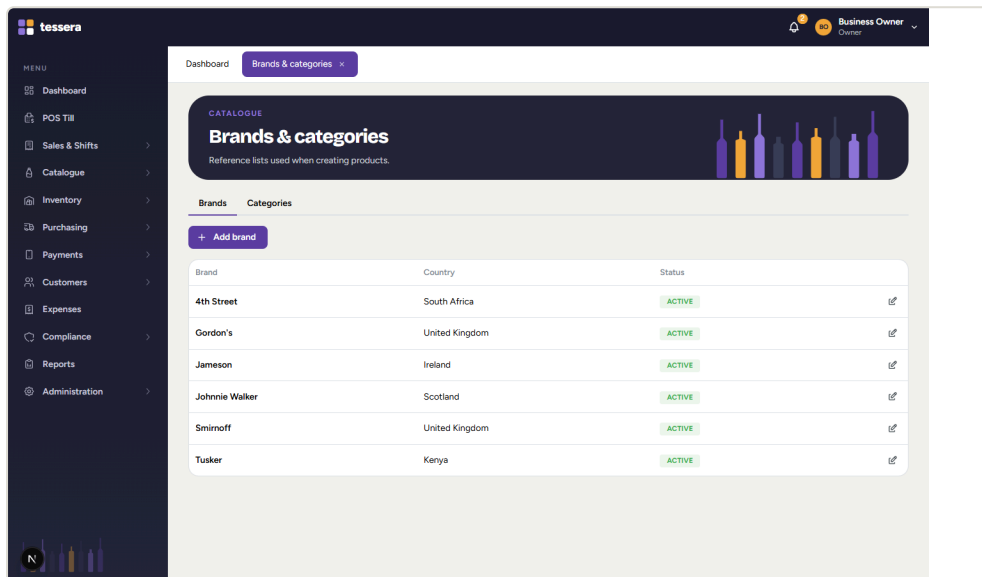


Figure 10: Catalogue → Brands & categories

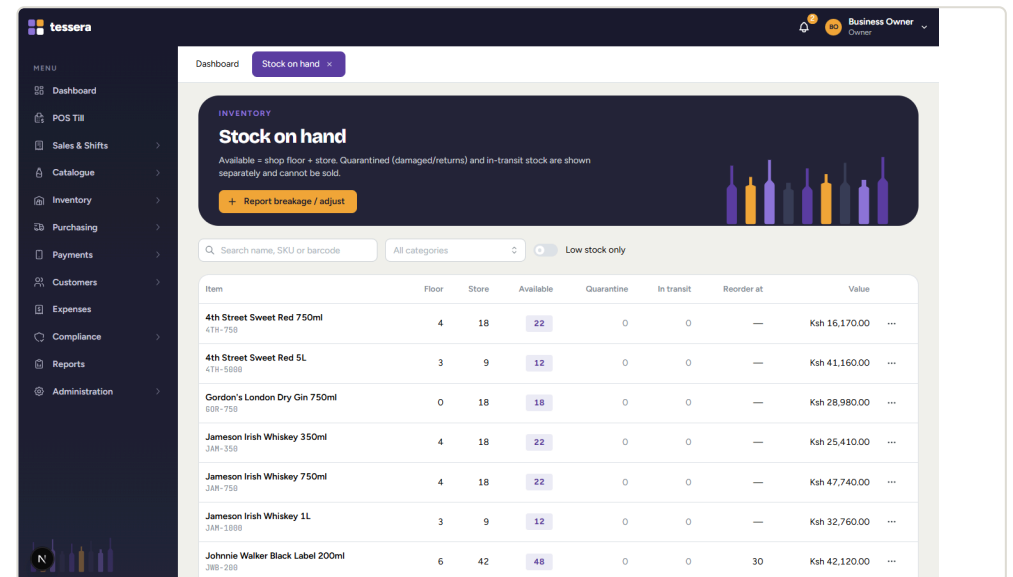


Figure 11: Inventory → Stock on hand

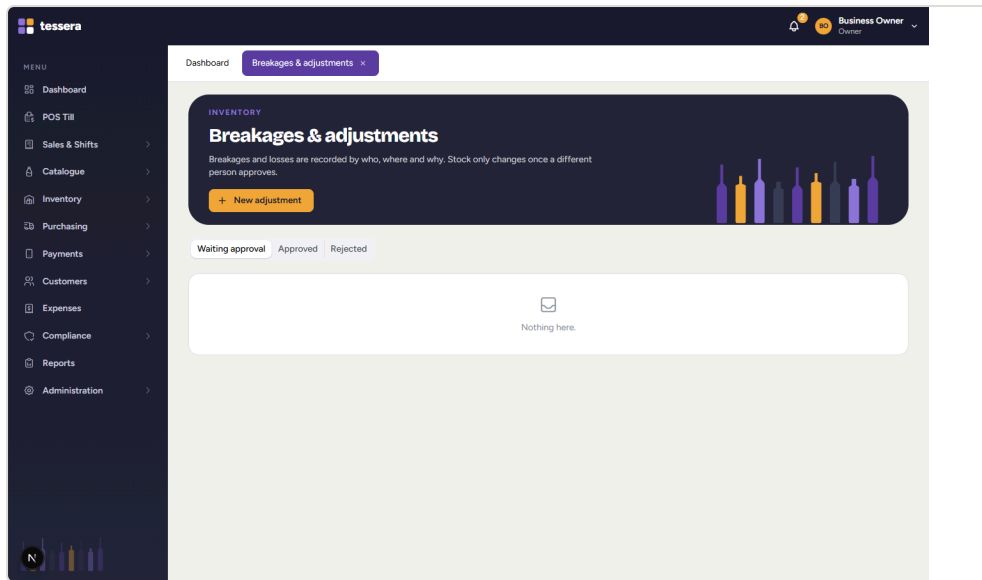


Figure 12: Inventory → Breakages & adjustments

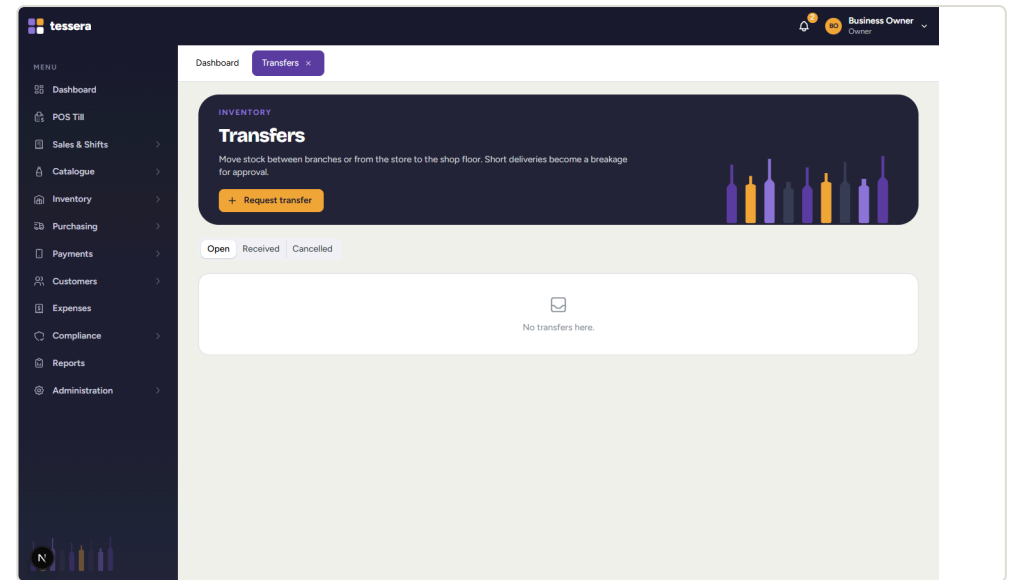


Figure 13: Inventory → Transfers

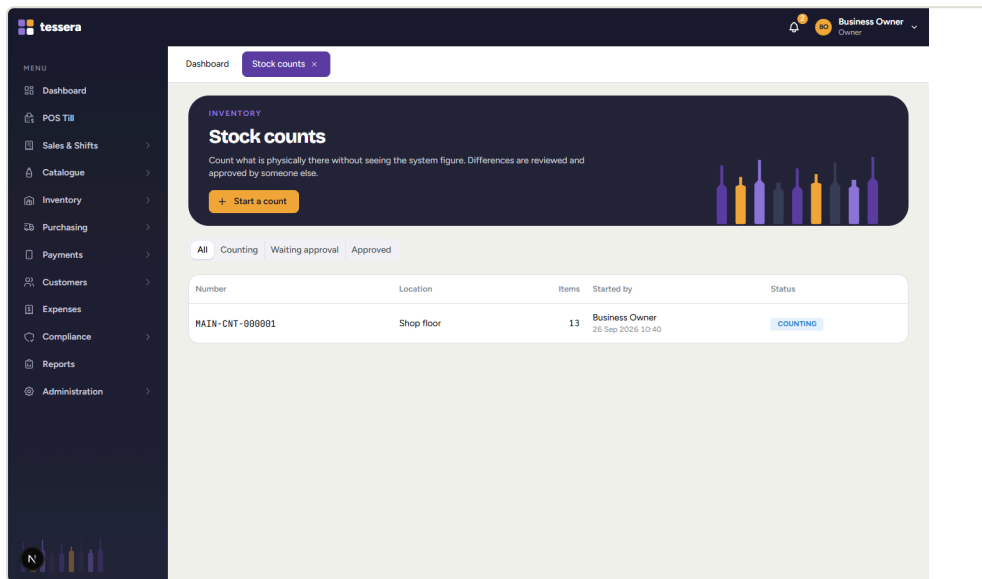


Figure 14: Inventory → Stock counts

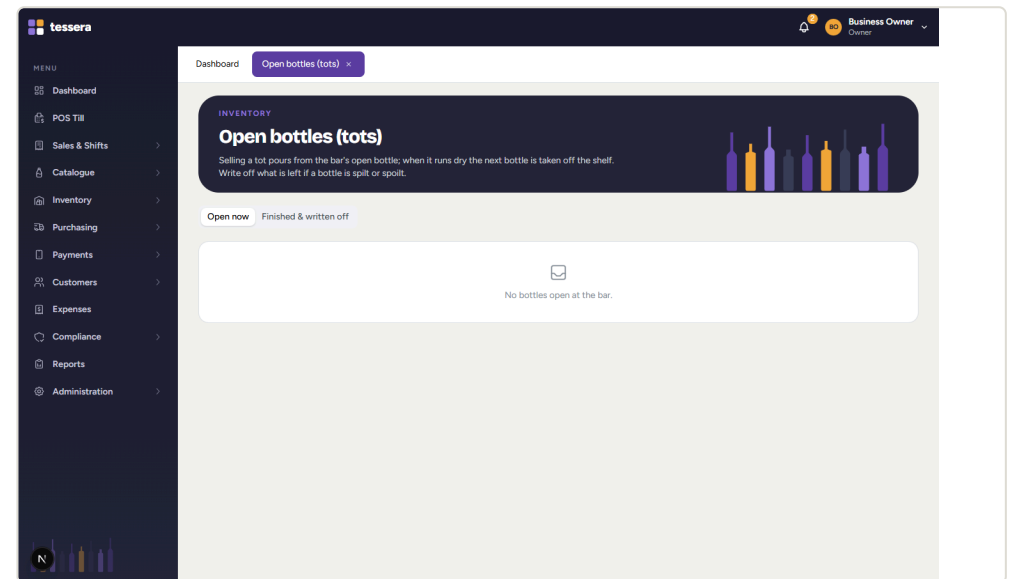


Figure 15: Inventory → Open bottles (tots)

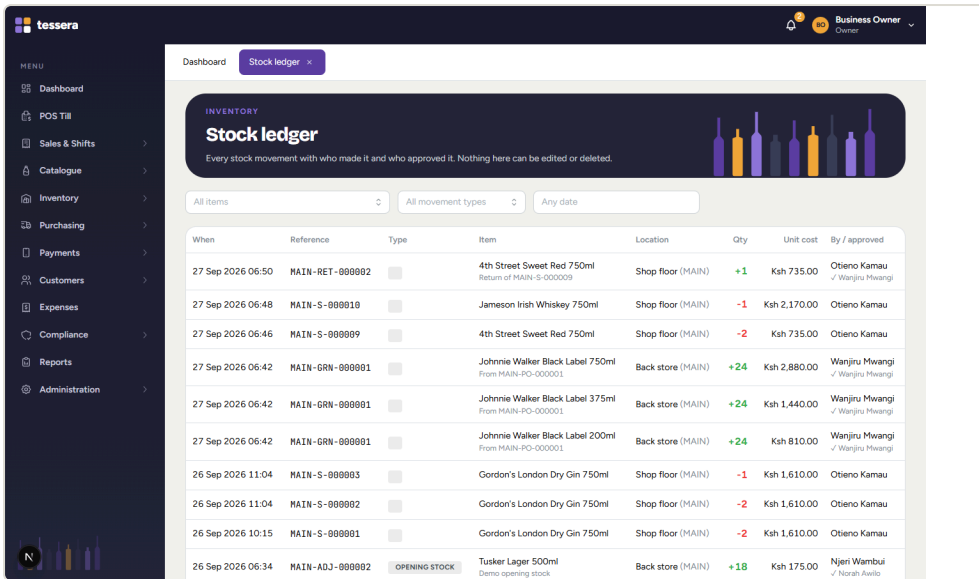


Figure 16: Inventory → Stock ledger

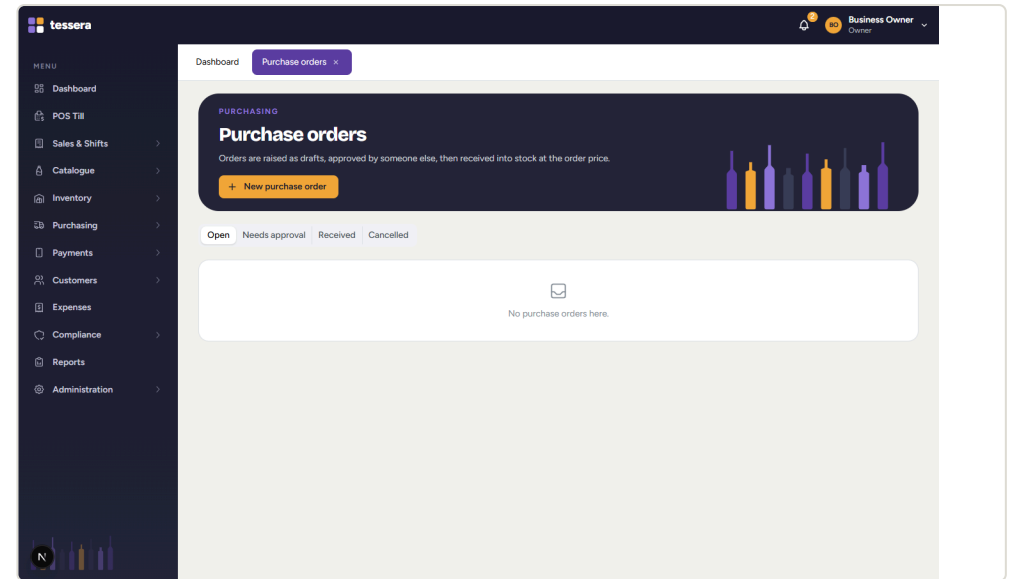


Figure 17: Purchasing → Purchase orders

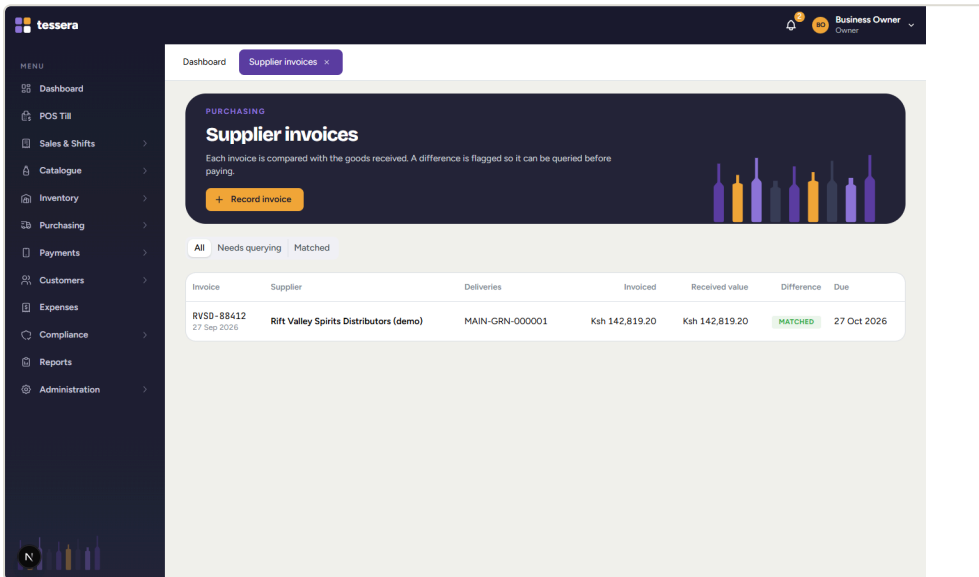


Figure 18: Purchasing → Supplier invoices

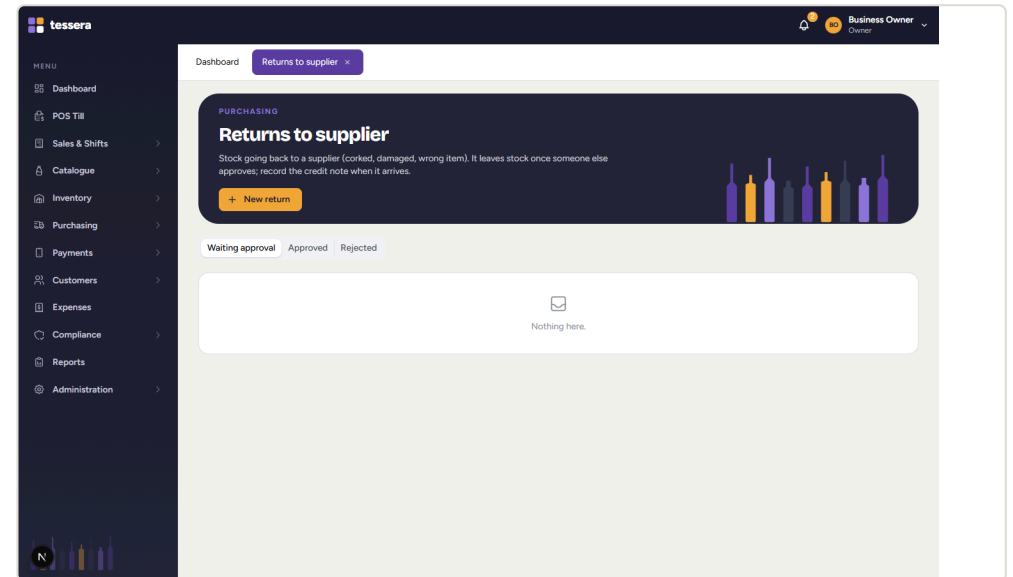


Figure 19: Purchasing → Returns to supplier

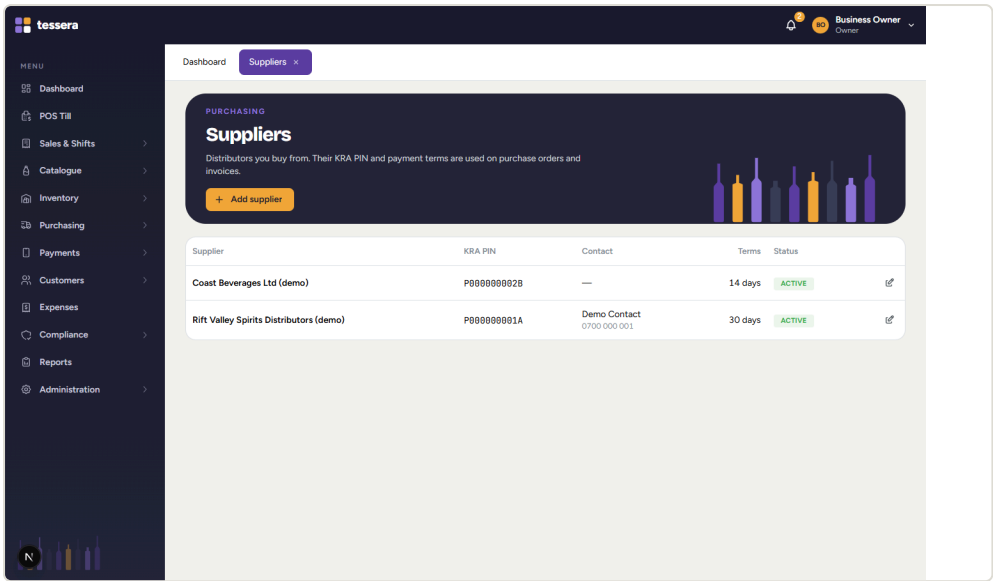


Figure 20: Purchasing → Suppliers

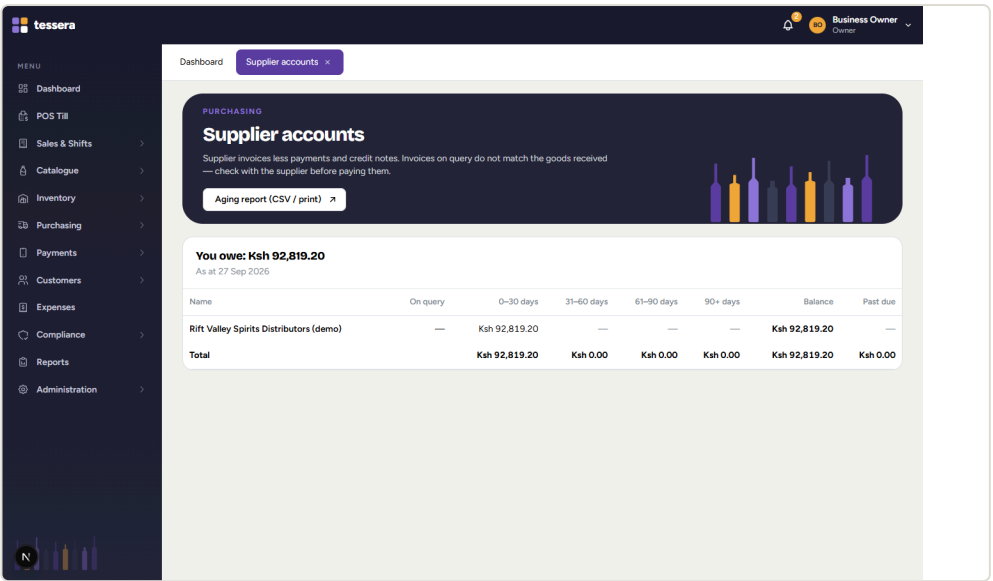


Figure 21: Purchasing → Supplier accounts

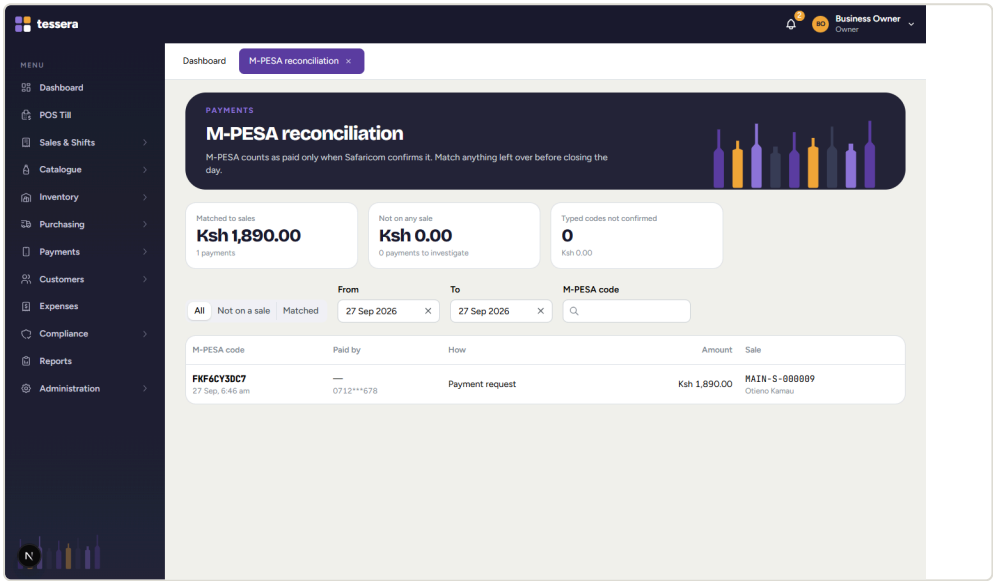


Figure 22: Payments → M-PESA reconciliation

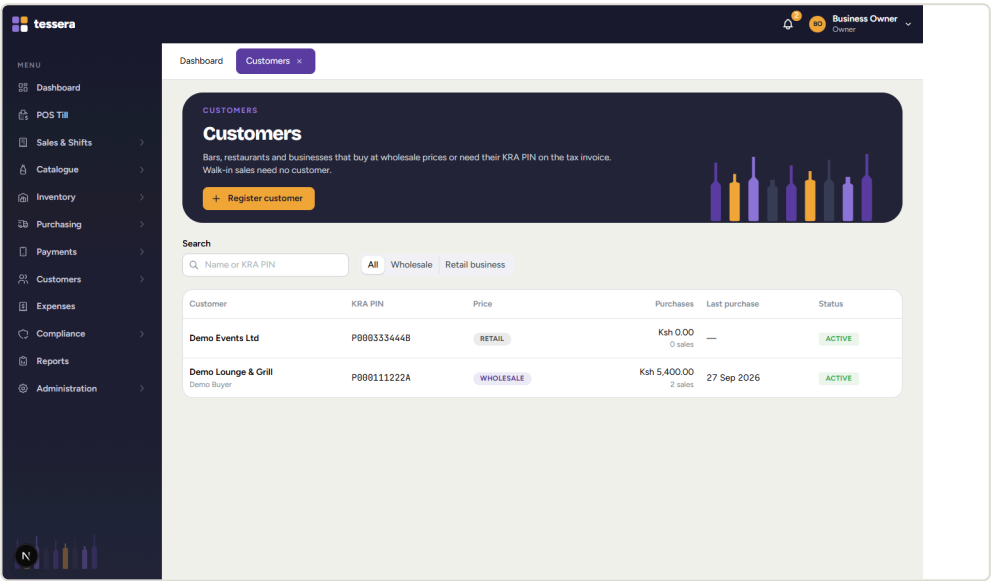


Figure 23: Customers → Customers

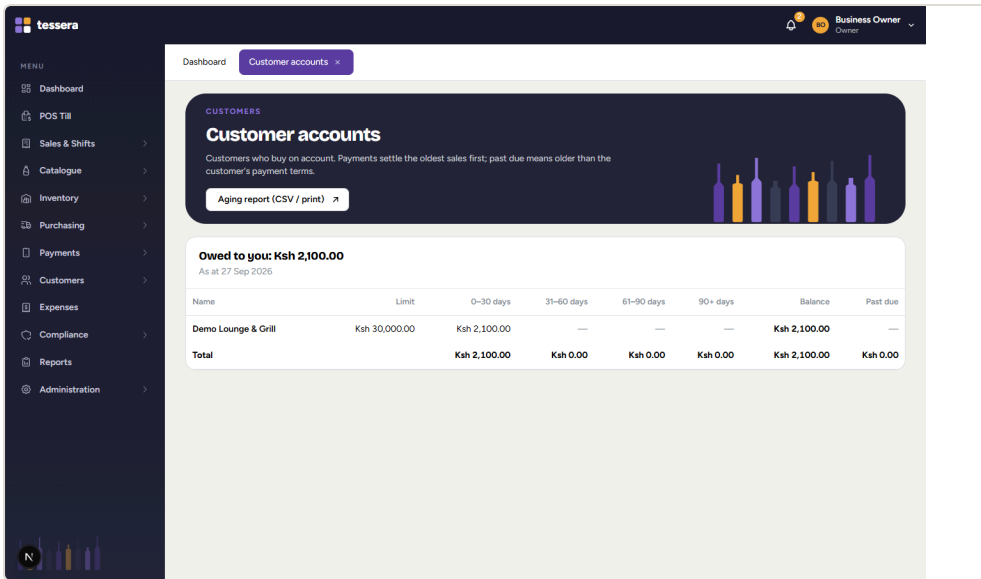


Figure 24: Customers → Customer accounts

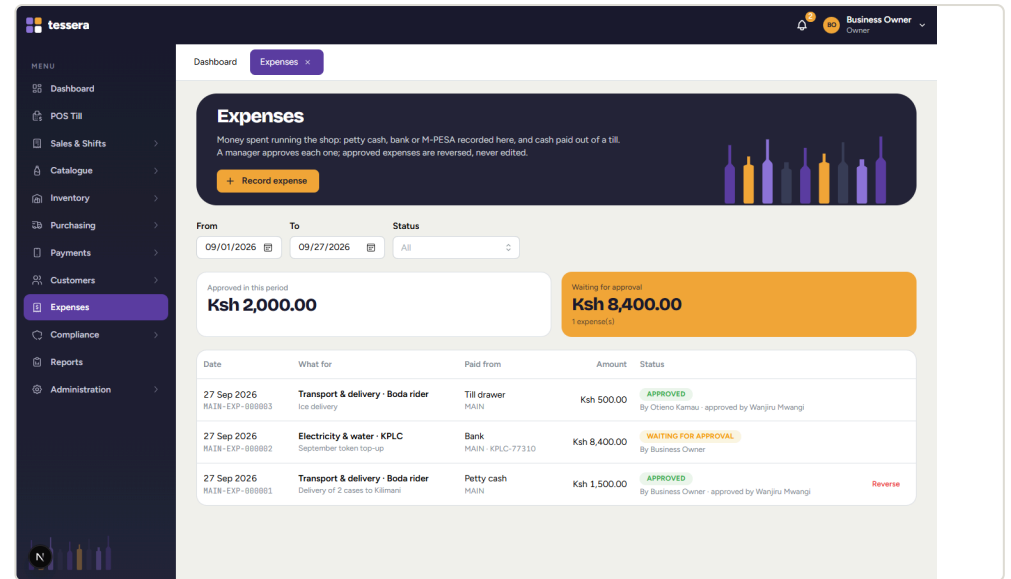


Figure 25: Expenses

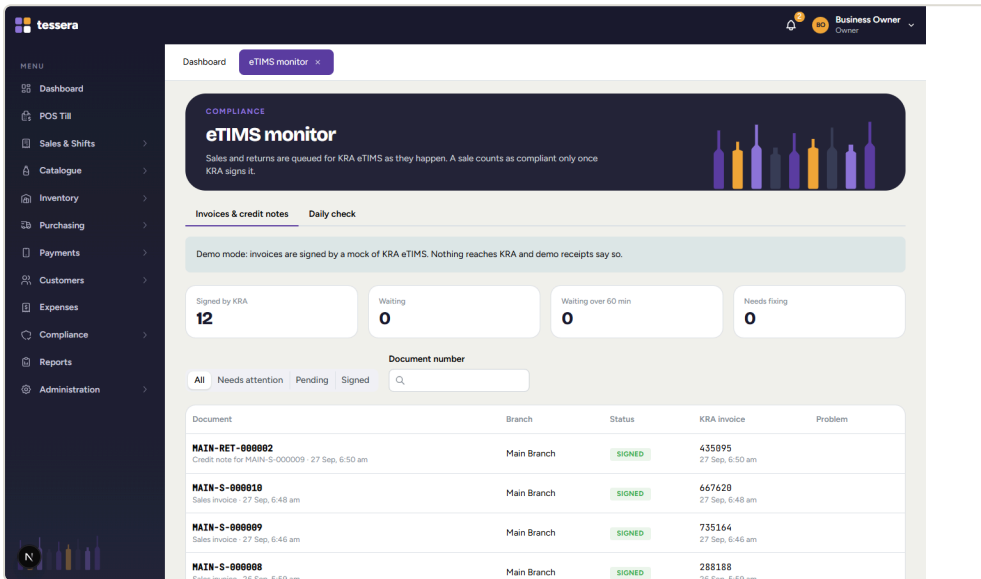


Figure 26: Compliance → eTIMS monitor

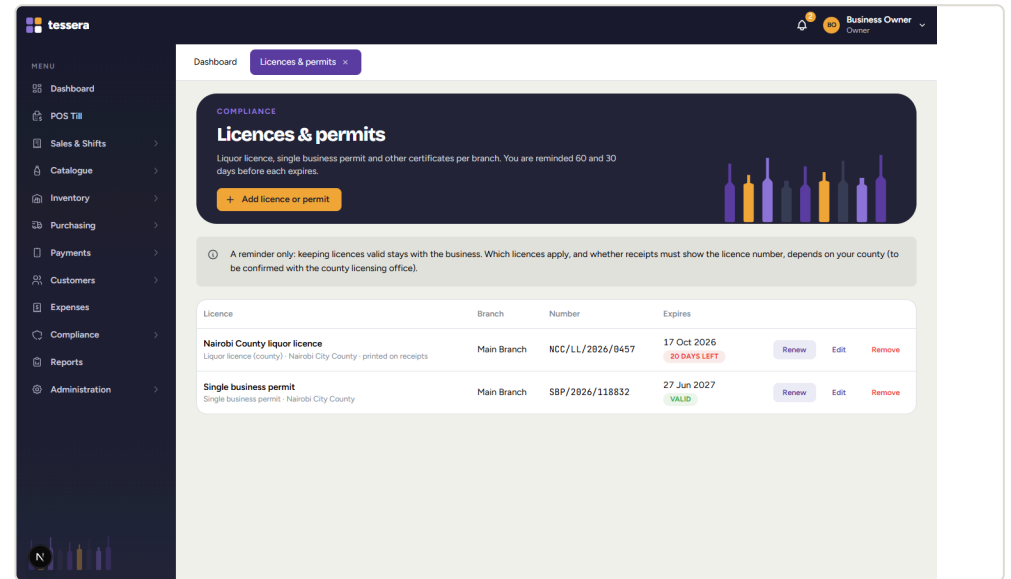


Figure 27: Compliance → Licences & permits

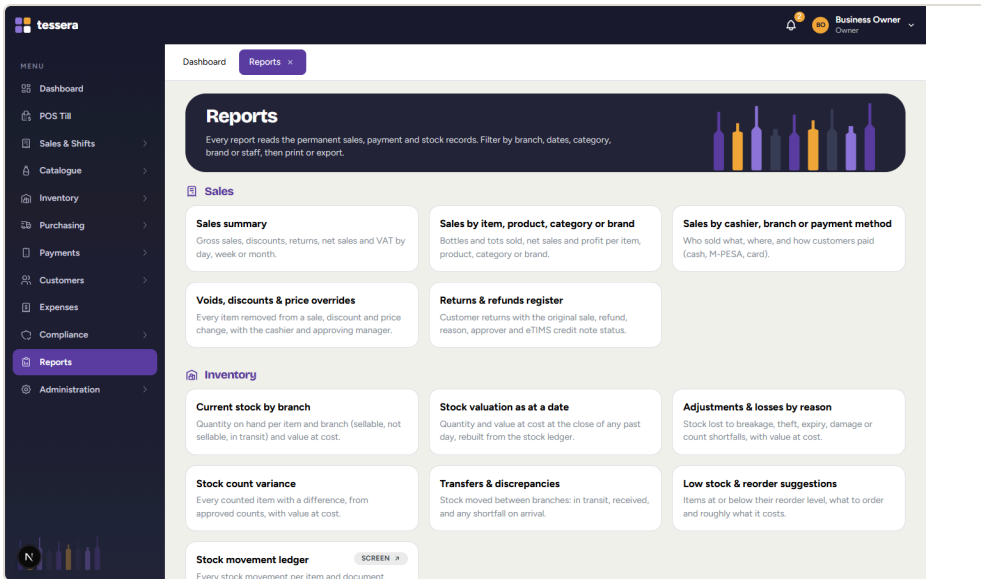


Figure 28: Reports

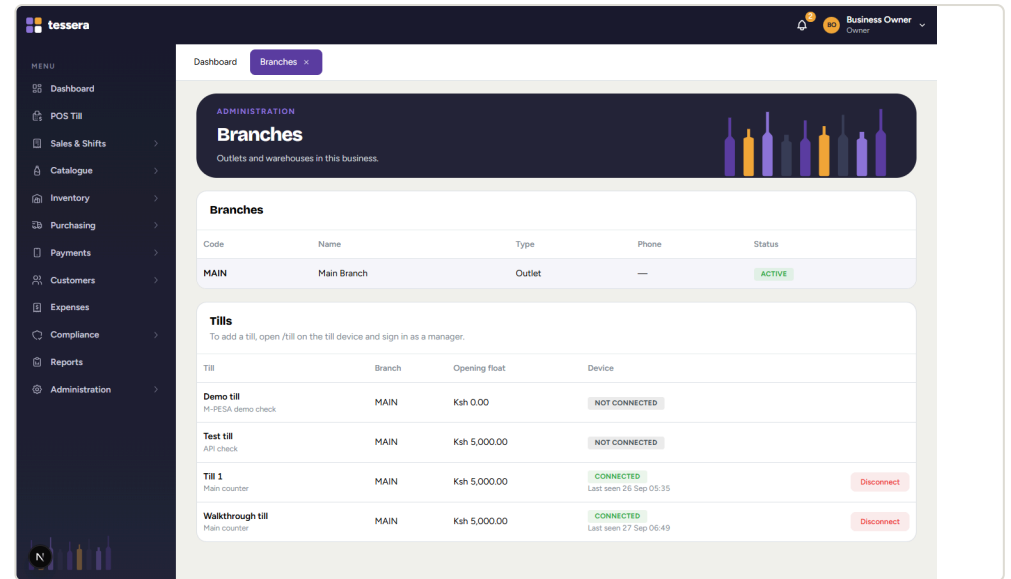


Figure 29: Administration → Branches

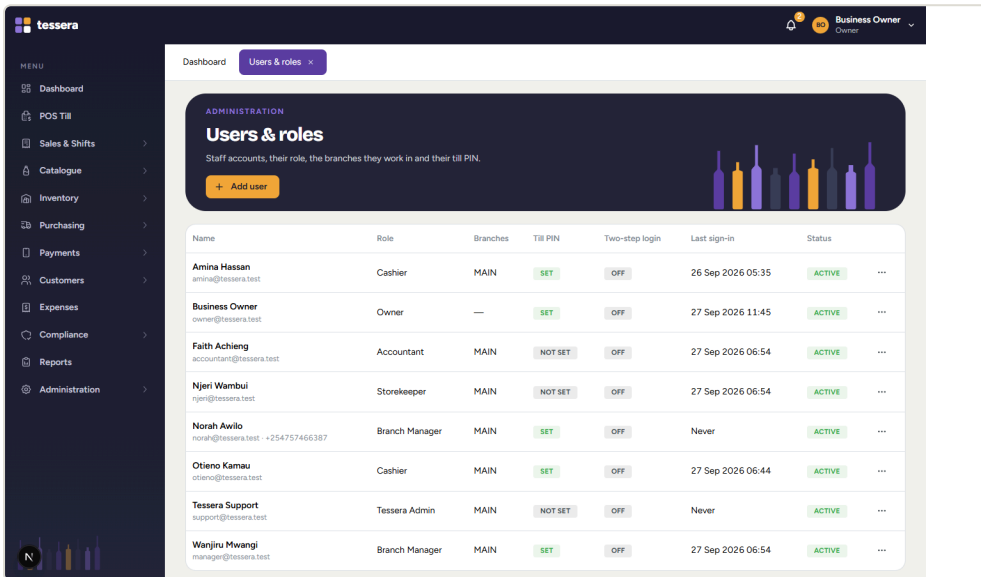


Figure 30: Administration → Users & roles

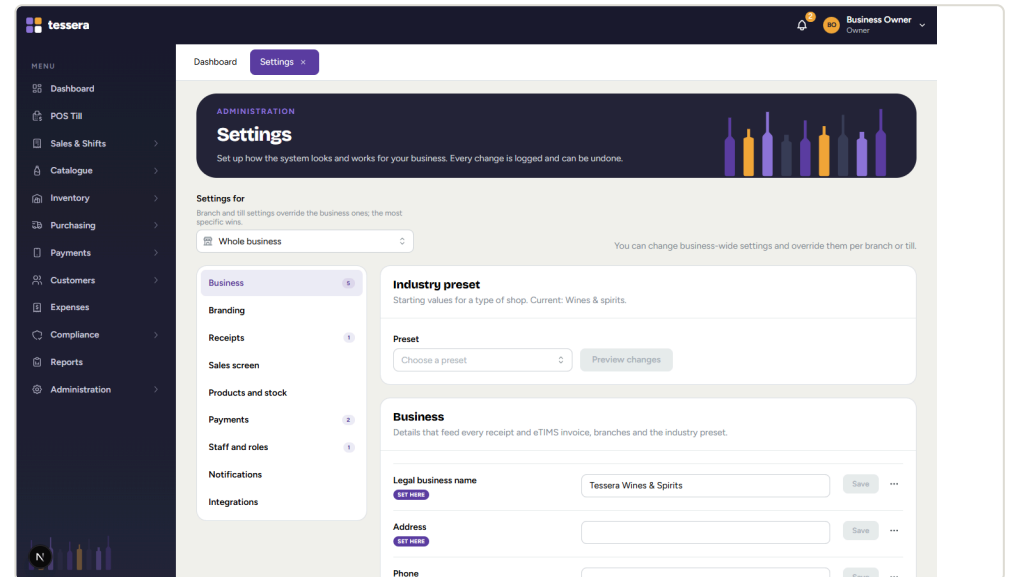


Figure 31: Administration → Settings

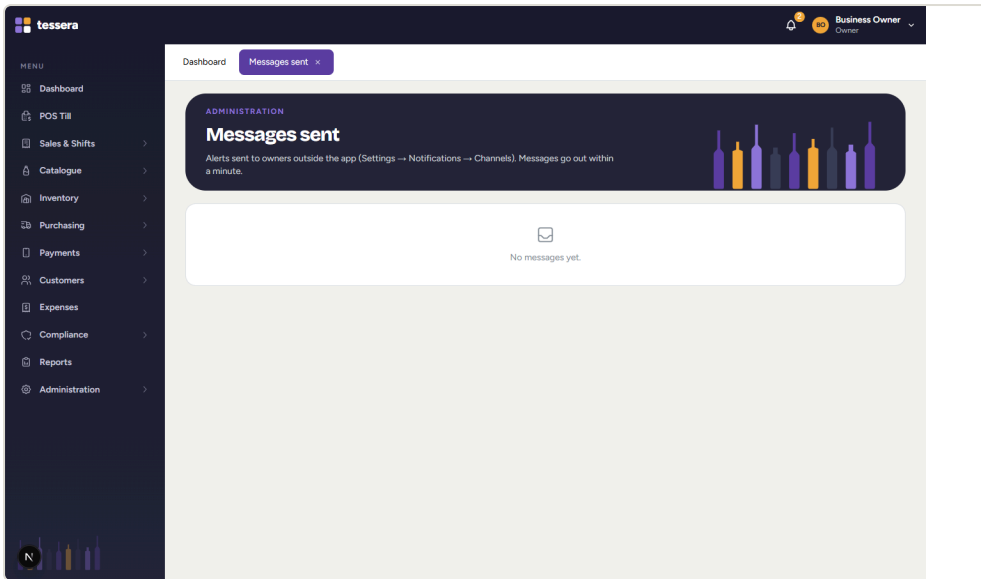


Figure 32: Administration → Messages sent

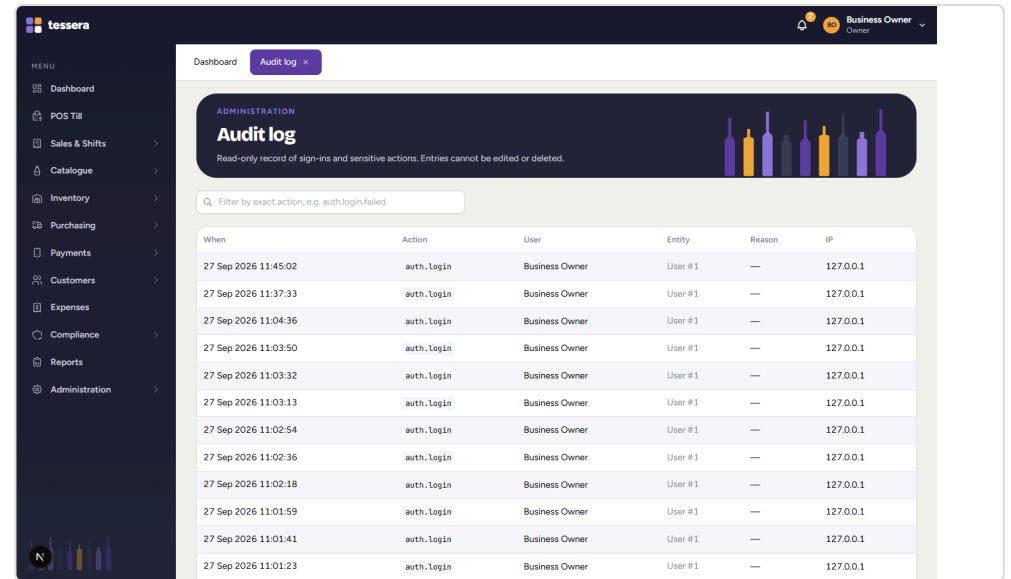


Figure 33: Administration → Audit log

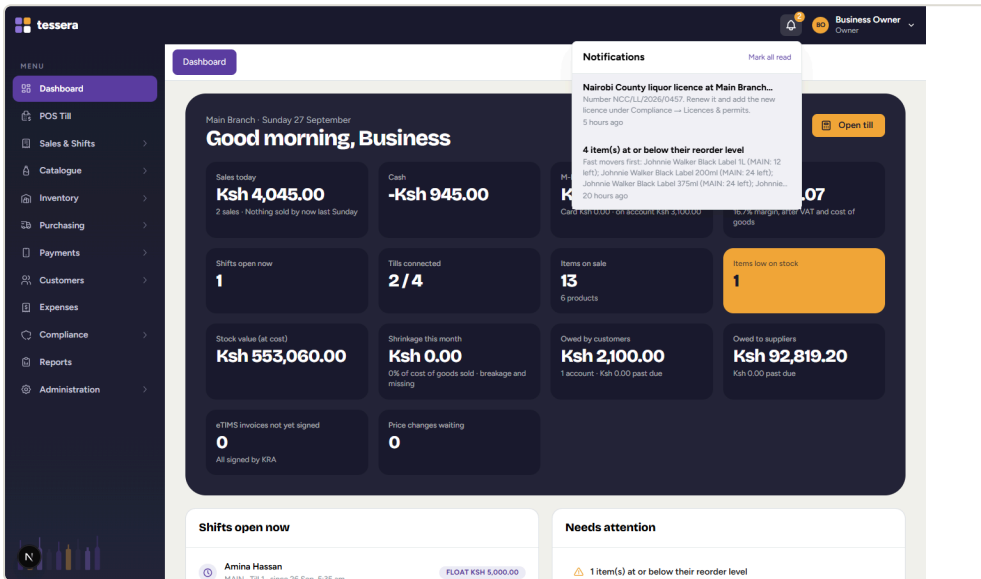


Figure 34: Notifications (the bell, top right)

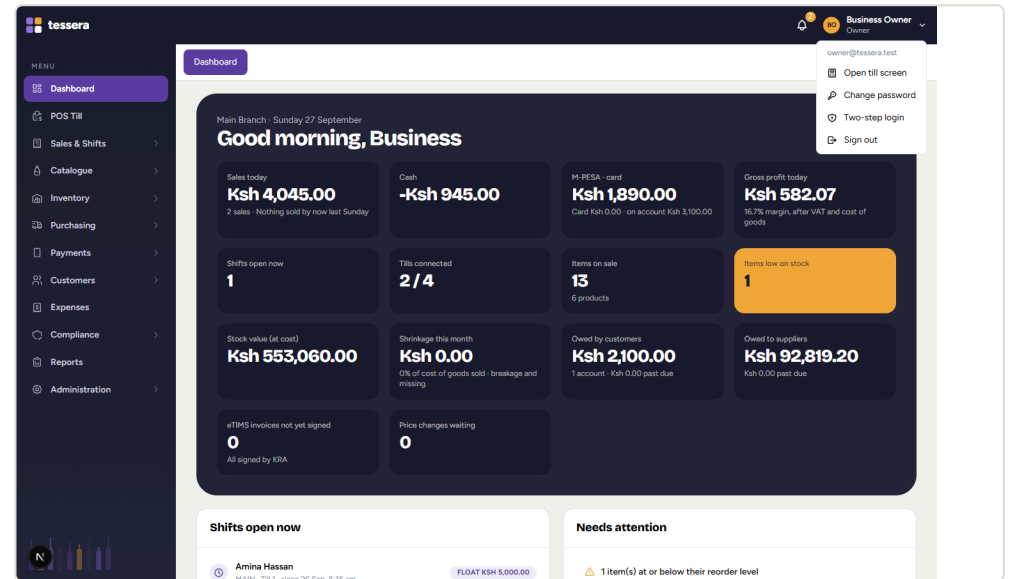


Figure 35: Account menu (top right)

5.2 The till

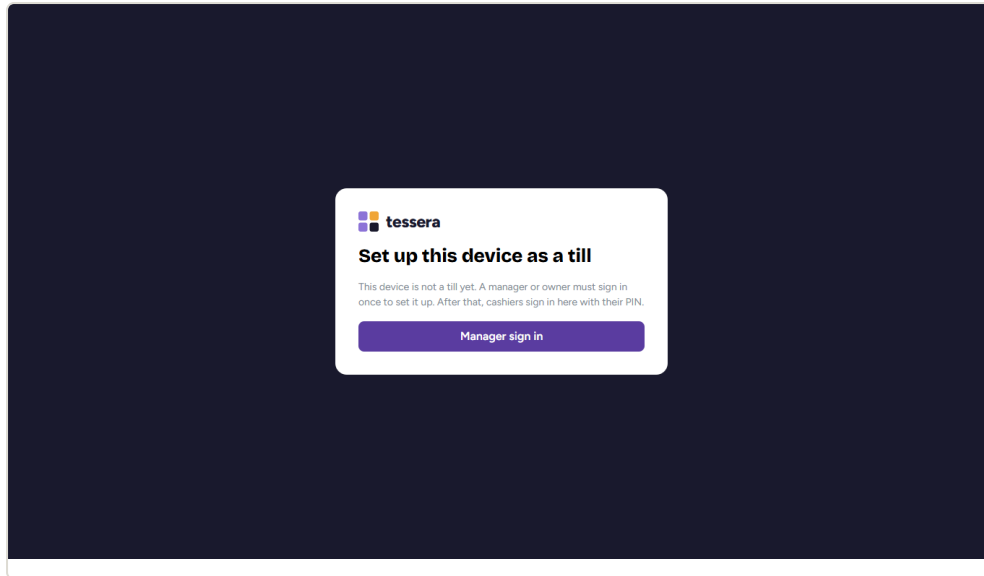


Figure 36: A computer that is not yet a till

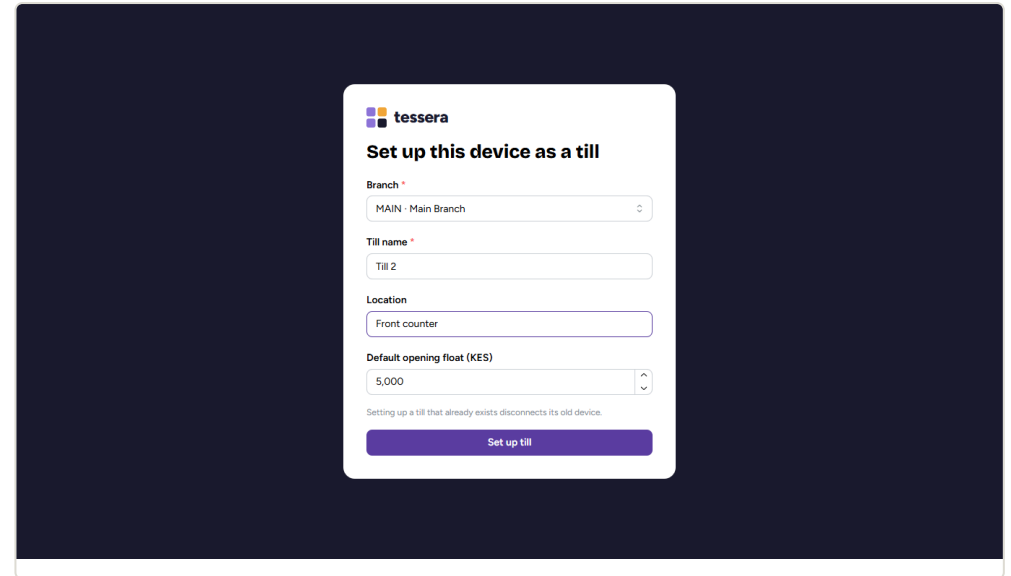


Figure 37: Setting up the computer as a till (manager signed in)

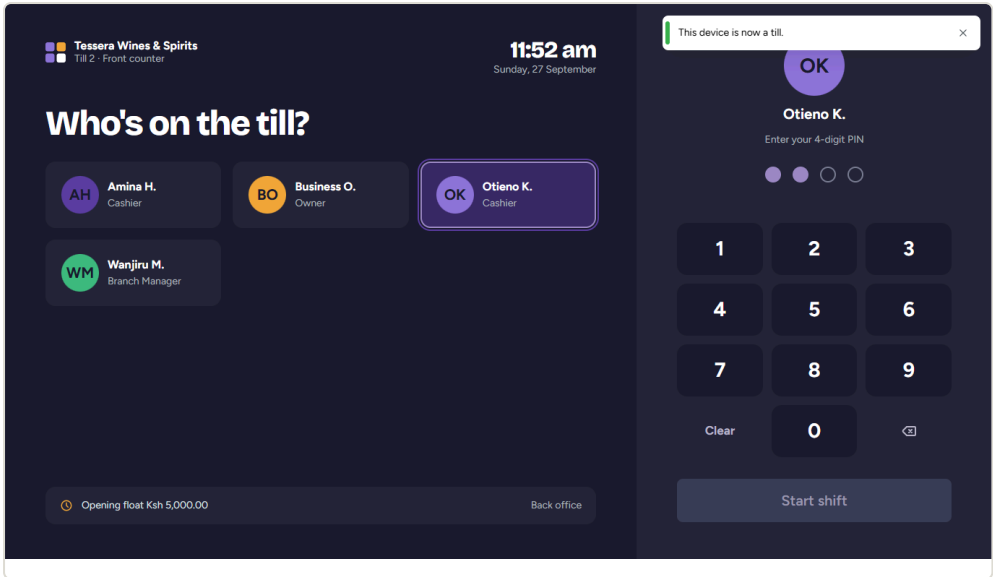


Figure 38: Who's on the till? The cashier enters a PIN

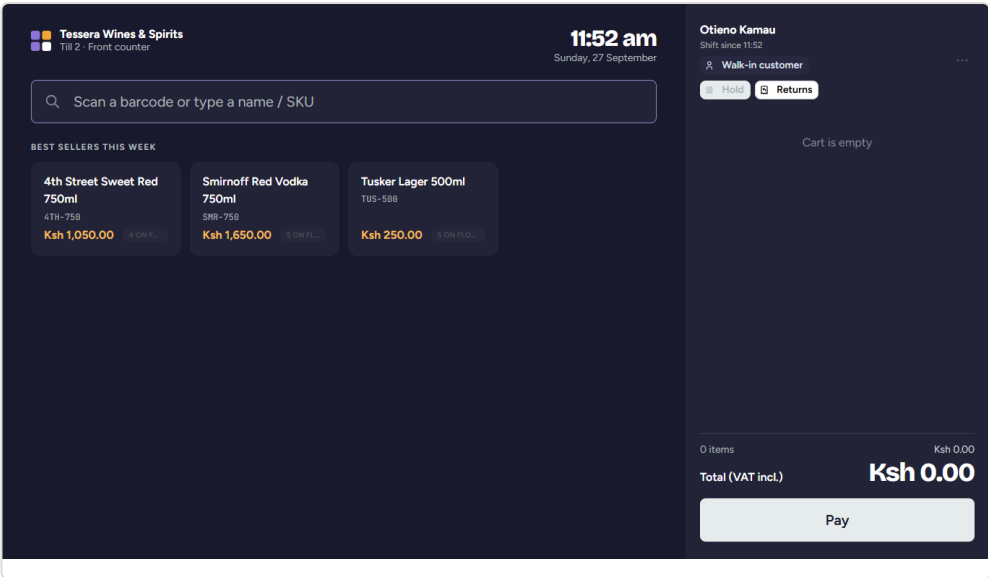


Figure 39: Selling screen with this week's best sellers

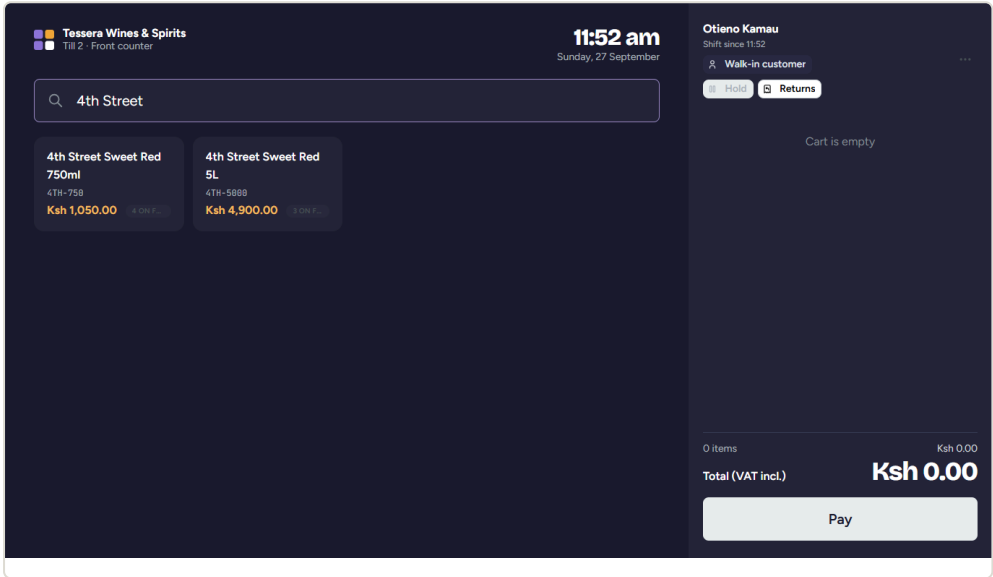


Figure 40: Searching for a product

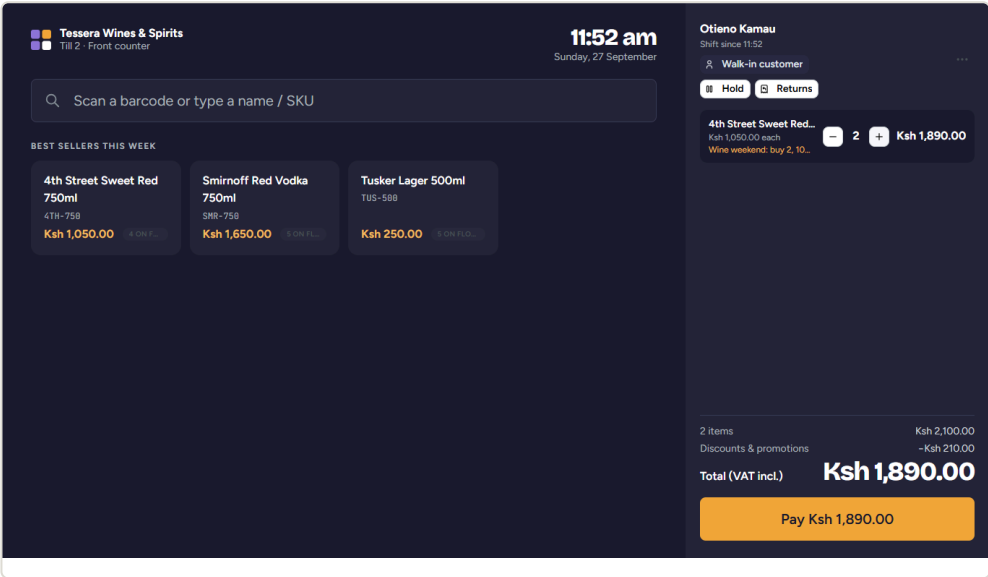


Figure 41: Two bottles of wine with the promotion applied

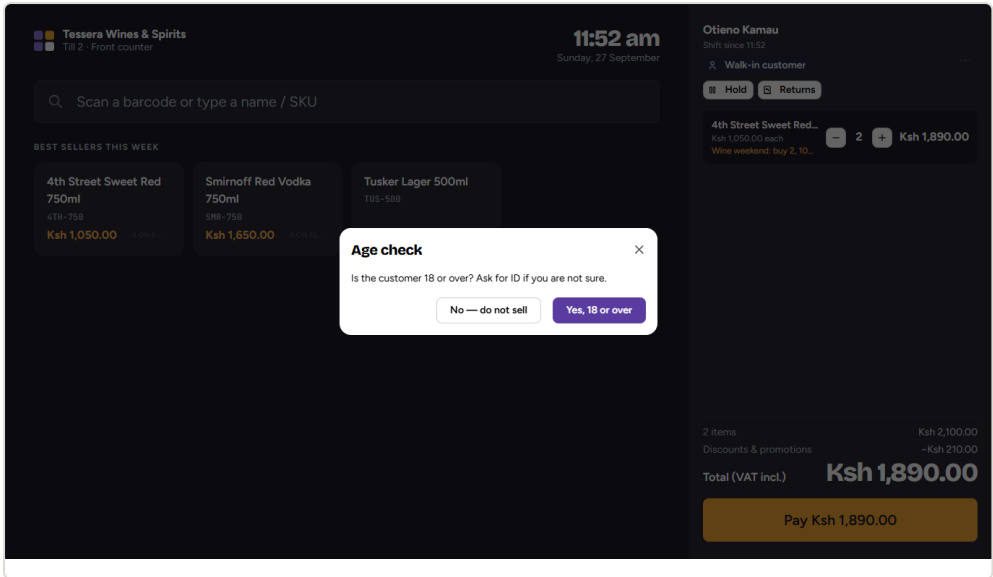


Figure 42: Age check before payment

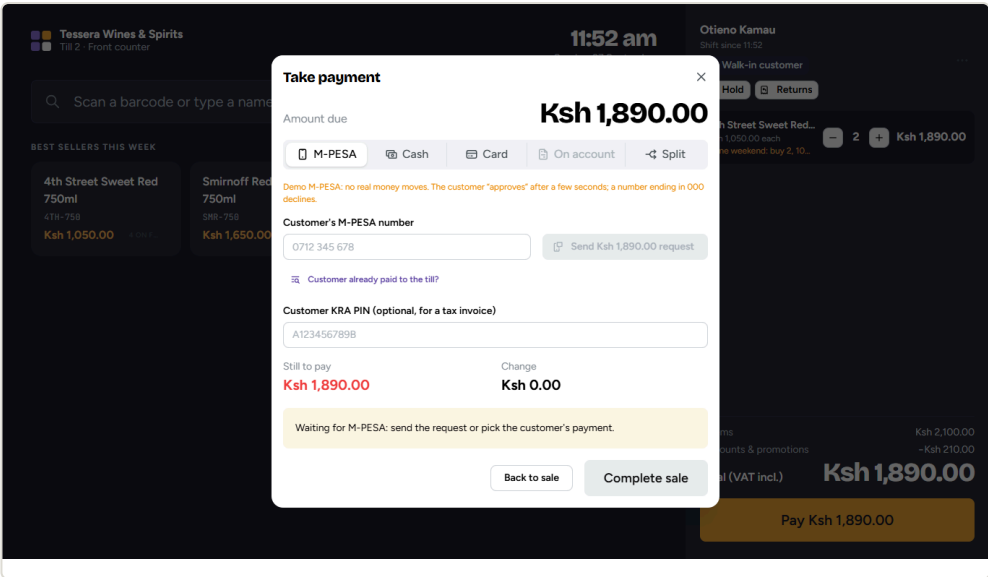


Figure 43: Take payment: M-PESA (demo)

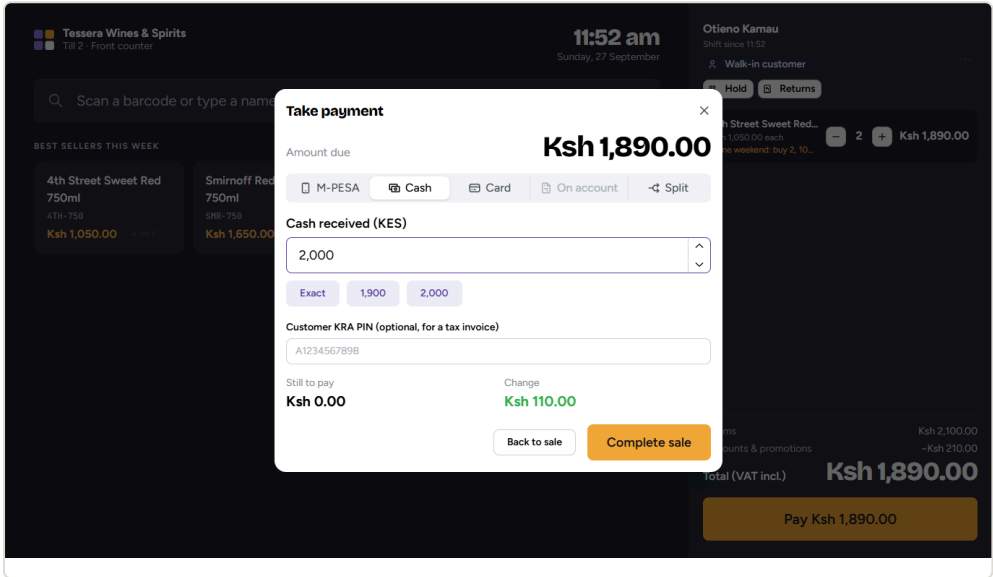


Figure 44: Take payment: cash, with the change worked out

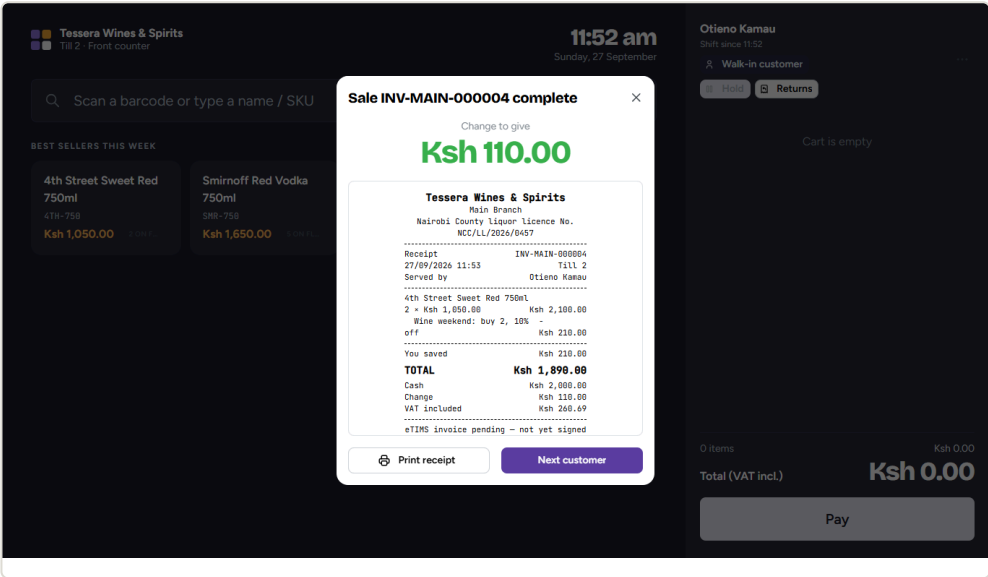


Figure 45: Sale complete: change to give and the receipt

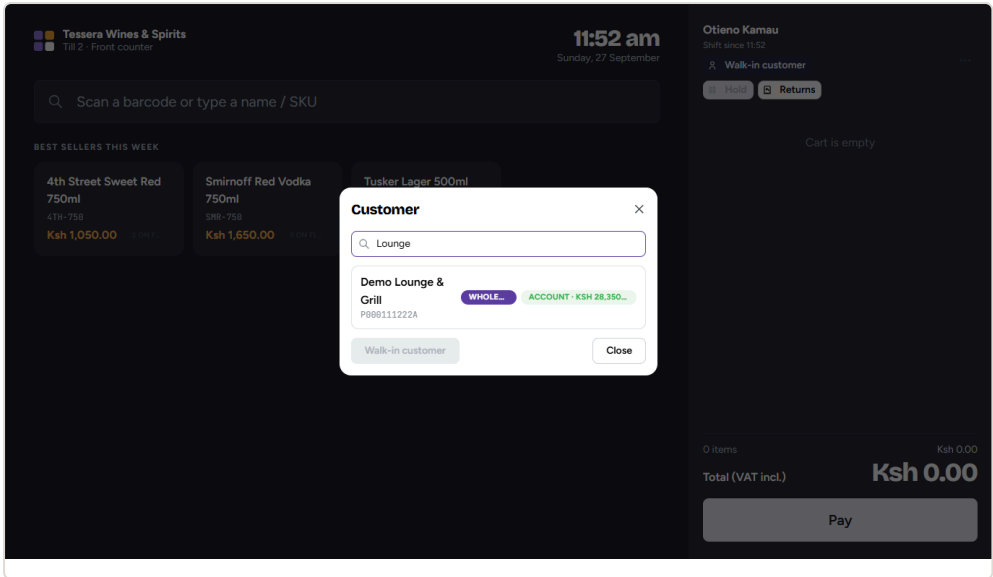


Figure 46: Choosing a registered customer

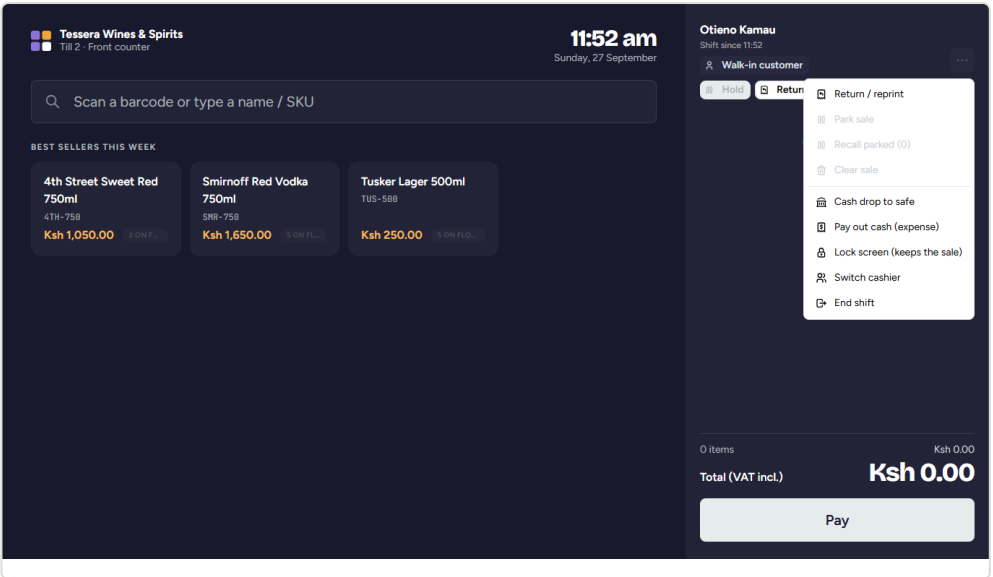


Figure 47: The till menu

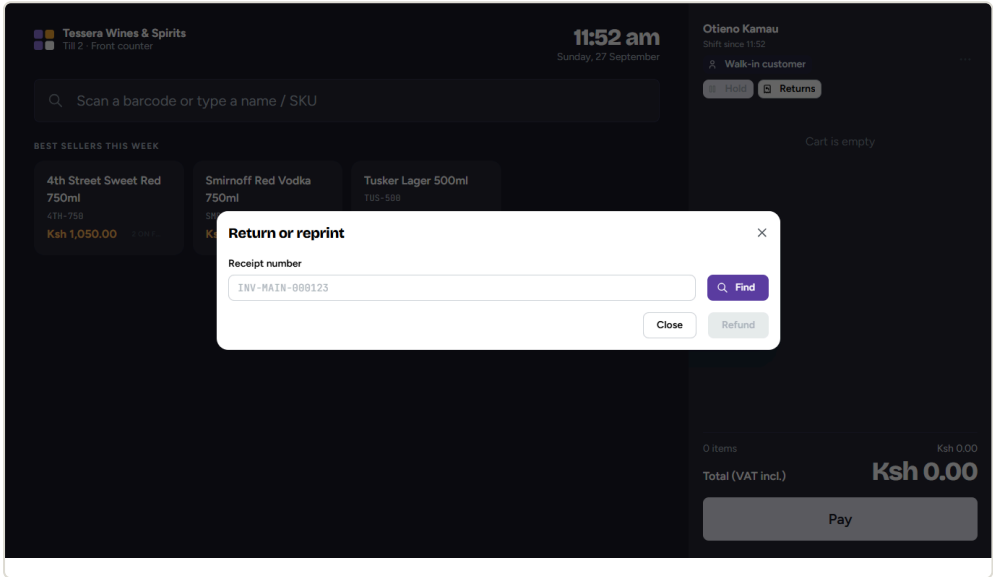


Figure 48: Return or reprint

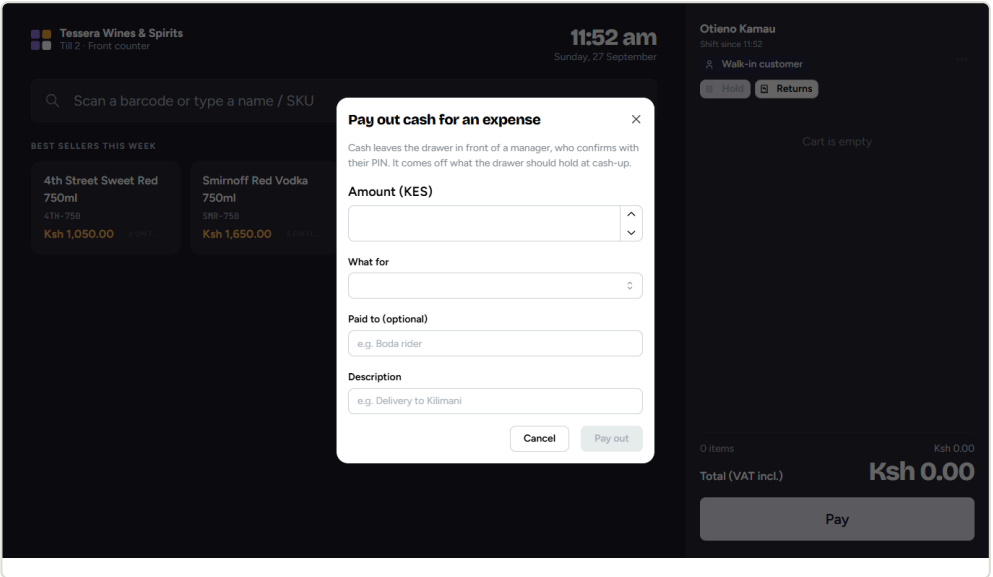


Figure 49: Pay out cash for an expense

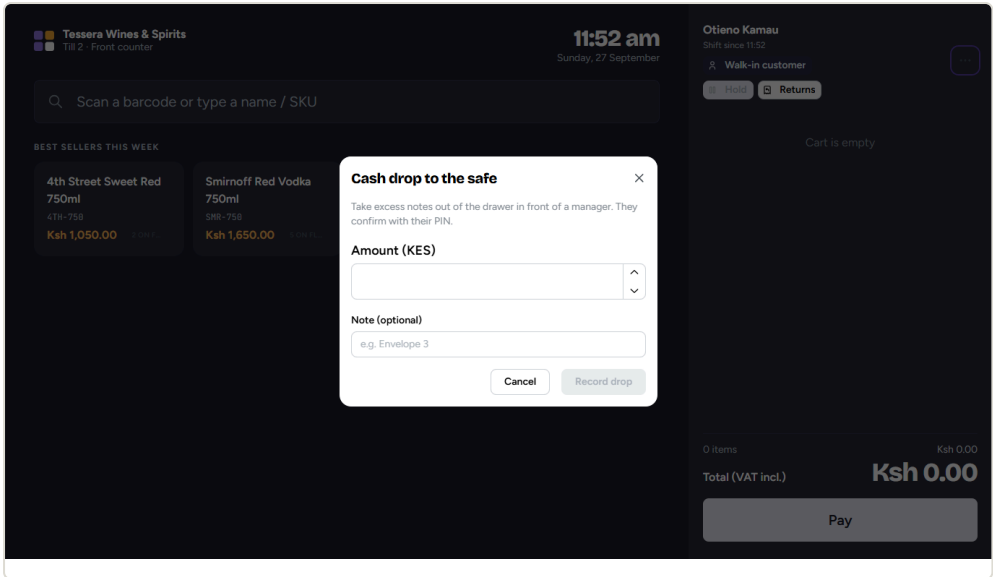


Figure 50: Cash drop to the safe

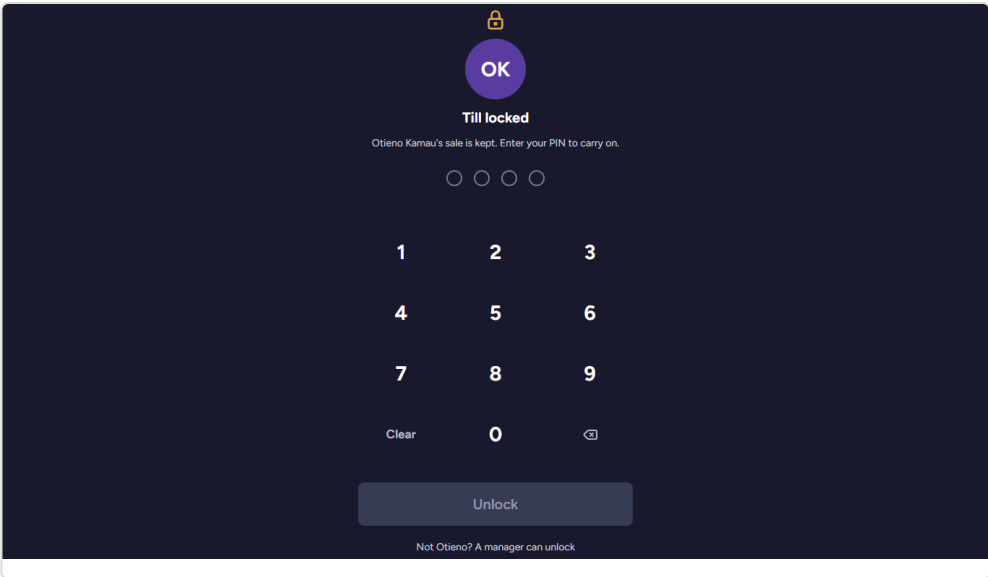


Figure 51: Till locked (the sale is kept)

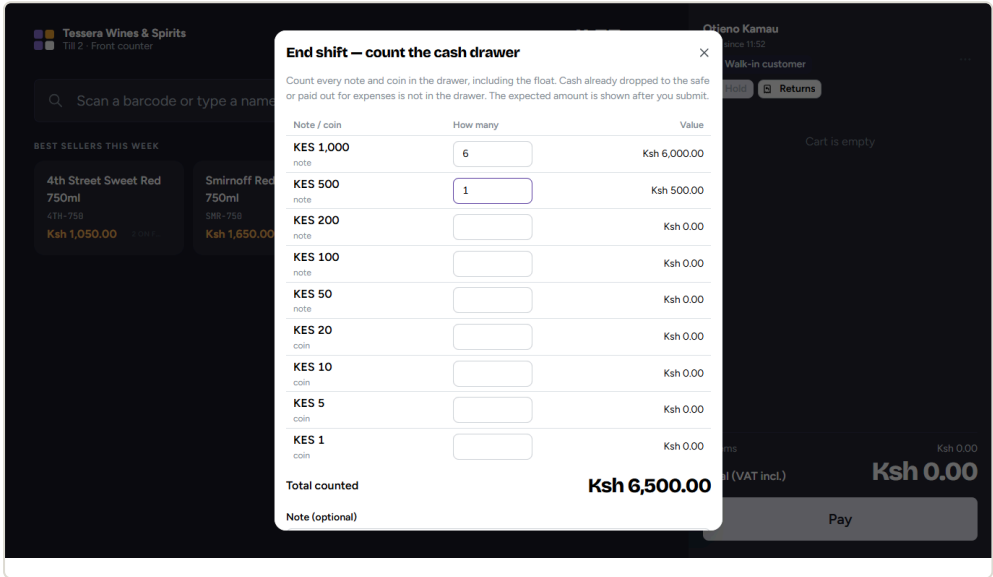


Figure 52: End shift: counting the cash drawer

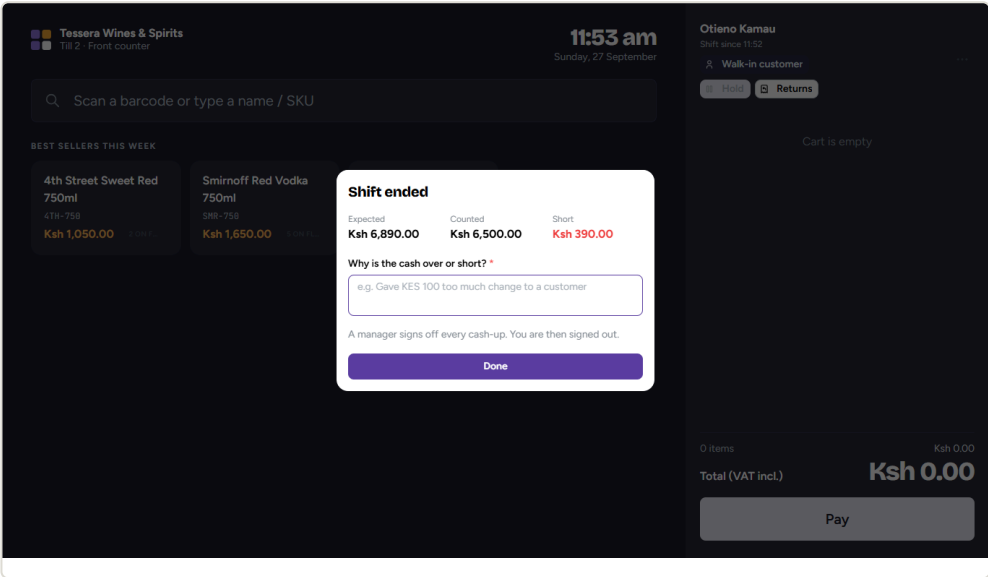


Figure 53: Shift ended: expected, counted and difference

6. Features in the brief that do not exist (will not be documented)

Brief item	Finding
Change language	Not built: the system is English only.
Reset password by yourself	Not built: "Forgot password?" tells you to ask the owner or an administrator, who sets a new password in Users & roles.
Receipts by SMS or email	Not built. The option has been hidden from Settings until it works (internal notes).
Expiry alerts, batch tracking	Not built. Batch tracking is hidden from Settings; "Expiry alerts" shows as Coming later.
X-report / Z-report	No reports with these names. The shift cash-up (end-of-shift count) and the "Shift cash-up & variance" report do this job.
Real M-PESA and KRA eTIMS	Built, but running on demo connections until the client supplies credentials. Screens show "Demo M-PESA" notes; no real money or tax data moves.
Phone / mobile app	None. The till is designed for a computer or tablet screen.

7. Decisions (27 September 2026)

#	Question	Decision
1	Approve the inventory	Approved. Write "How everything connects" first, then Sales/POS.
2	"SMS or email only" receipts and batch/expiry tracking	Hidden from Settings until they work. Not in the client guide; listed in <code>internal-notes.md</code> .
3	Cashiers and the Customers menu	Intended, with limited rights. Checked: cashiers can search and select but cannot add a customer (fix needed, internal notes), and cannot edit, delete, export or see balances (correct).
4	Storekeeper dashboard	Their home screen. The missing menu item was a bug and has been fixed (Dashboard added to the storekeeper menu).
5	Who raises purchase orders	Document the current permissions (owner, admin or accountant). Storekeeper requests are "coming soon" (internal notes). The demo PDF has been corrected.
6	Managers and prices	Intended: prices are set centrally. The guide explains why, and that managers still approve till discounts up to their limit.
7	"Coming later" settings	Mentioned only as not available yet.
8	Screenshot sizes	Back office 1440×900, till 1366×768.
9	Demo data	Separate guide database, "Karen Wines & Spirits" (Karen and Westlands branches).
10	Output	Word and PDF from the Markdown source.